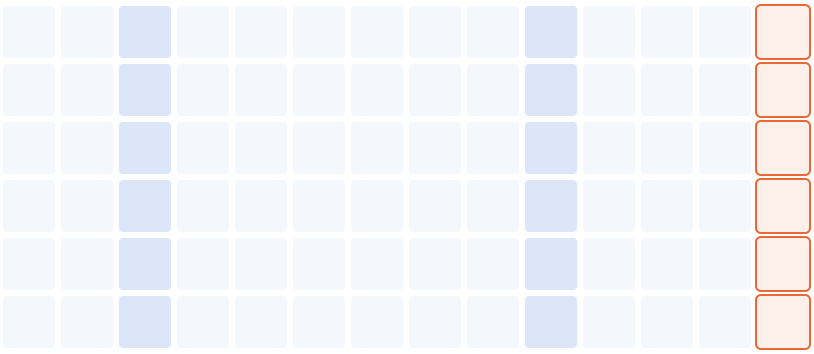
RESEARCH PREPRINT

The D'Agapeyeff Cipher (1939)

Grille recovery and polyphonic-table identifiability under n-gram scoring



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1. Executive summary

Publication status. Research preprint. This manuscript has undergone internal verification and AI-assisted critical review but has not been formally peer-reviewed by an independent academic journal.

In 1939 Alexander D'Agapeyeff closed his book *Codes and Ciphers* with a short challenge cipher: 196 two-digit numbers, and invited readers to solve it. In more than eighty years nobody has.

This report sets out what an AI-assisted cryptanalysis program, conceived and directed by Nicolas Houghtaling over seven days, October 8 to 14, 2026, established about the cipher, with further checks on October 15 to 17 prompted by reviews of the draft by a separate AI system, ChatGPT (OpenAI). It is careful to separate this study's findings from what others published first, and what was measured in practice ciphers from what can be said about the real one.

The question addressed. Recent work (Hendrick 2026) showed that no one-to-one letter key, combined with any transposition, can produce the cipher's symbol counts, and that if the cipher carries a message at all, its order was scrambled by a system as thorough as a *turning grille* (other families, such as keyed four-square, also remain open). One construction consistent with every observed property is a 14 × 14 turning grille laid over a table in which each of 13 symbols stands for a group of letters (a *polyphonic* table). Hendrick left the turning grille open because no search could find its key. We asked whether, under this hypothesis, it could be found at all.

What we found. All rates below are measured on practice ciphers we made in the hypothesized form, with known keys.

- **With the true table supplied, practice-cipher grilles were recovered.** Scoring candidate grilles by six-symbol patterns recovered at least 45 of 49 orbits in six of six practice ciphers (five exactly; 95% interval 61–100%); four-symbol scoring, the usual choice, managed 2 of 6. A preliminary demonstration on six synthetic ciphers with the table given — not a general success rate, and not recovery of the historical grille.
- **A nearly right table is not enough.** With 6 or more letters in the wrong group the search never found the grille (0 of 12); with the true table it did 4 of 4. A preliminary result, the polyphonic counterpart of Hendrick's for one-to-one keys.
- **Under the scores tested, 182 symbols rarely single out the table.** Handed the true symbol order and started *at the true table*, a search that climbs the n-gram score usually moves to a *different table that scores higher*. The true table stayed best among its neighbors in only 11% of 200 practice ciphers at the cipher's 182 symbols, and in 8% to 26% under every alternative tested after review: natural prose instead of sampled words, three English corpora and four scoring methods. It reaches one in two only at several hundred symbols.
- **The higher-scoring tables fit the passage, not English.** They usually still win under an independent corpus, but on a fresh passage enciphered with the true table they lose four times in five: they are fitted to the particular 182 symbols. A higher score shows the true table is not optimal under the score, not that another table is more probable under English itself. Checks of the scoring itself found no sign that it favors particular tables: on 10,000 and 100,000 held-out symbols, no tested neighbor outscored the true table.
- **D'Agapeyeff's own book teaches no table of this kind.** Every construction in it puts one letter in each cell.
- **A second construction can also fit.** Four-square and two-square with *random* keys spread English over 18 to 25 symbols: in 10,000 practice ciphers none reached the cipher's 13, which excludes randomly keyed squares for the

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plaintext tested, not every key. But a four-square whose key is built on purpose around the message reached 13 symbols, as flat as the cipher's, in 13 of 100 passages; 10 of those 13 were not flagged by the family-wise order test used in this study at the 5% level, which does not show that they contain no order. That shows such a construction is possible, not that D'Agapeyeff used it; it has not been searched.

What it means. Under the grille-plus-polyphonic hypothesis and the n-gram scores we tested, 182 symbols are not enough to single out the table, so with these methods the hypothesis can be neither confirmed nor refuted from the ciphertext alone. This is a measurement under stated methods, not a proof that no method could succeed. For the tested scoring objectives, higher-scoring incorrect tables frequently exist even when the true symbol order and table are supplied, so additional optimization of the same objective cannot be assumed to improve true-table recovery; we have not established how success rates change with longer search budgets. A differently designed search or a stronger model of the language is not ruled out. What would settle it is the table from a source outside the ciphertext, or a longer message in the same system. We claim no decipherment.

What this study adds, within the literature reviewed. Two results extend earlier work to polyphonic tables: recovery of the grille when the table is known, and the requirement that the table be essentially exact (Hendrick showed both for one-to-one keys). Three are new findings within the literature reviewed: the truth-start measurement and its robustness checks, the exclusion of randomly keyed four-square and two-square by symbol usage, and the feasibility of a purpose-built four-square. They are measurements and applications made with standard search and statistical tools, not new algorithms (Appendix A), and the literature check of October 14, 2026 behind them does not establish priority (Section 9). The flatness of the symbol counts, the count bound, the grille search with a known key and the one-to-one key tolerance are prior work, credited in Section 9.

Every number in this report comes from a recorded result: the files behind the key results are in the accompanying repository, listed with their SHA-256 fingerprints in Appendix C, and every other figure comes from an entry in the project's case file (Appendix D) that records its evidence. Appendix A describes the techniques built or adapted in this study, Appendix B the approaches that did not work, and Appendix E every setup of the robustness checks.

Codes in brackets, such as [F78] or [R29], refer to entries in the project's case file of checked results; each is explained in one line in Appendix D, and in the PDF each code links to its entry.

2. The cipher

In 1939 Alexander D'Agapeyeff closed *Codes and Ciphers*, a popular introduction to cryptography, with a challenge cipher printed without key or method. As transcribed in every standard source, it is 395 digits: 78 groups of five plus a closing "92000" [F1]. Read as two-digit numbers, the first 392 digits give 196 symbols, and 196 is exactly 14 × 14.

Laid out 14 symbols to a row (Figure 1), the text shows its one clear structural feature. The eight occurrences of the rare symbols (71, 92, 93, 94 and the single 04) all fall in the last column. To judge how unusual that is without fixing the row length in advance, we scored every row length from 2 to 20 by the largest share of the eight in any one column, less the share chance alone would put there (one in the number of columns), and kept the best score. None of 20,000 random placements of eight symbols among the 196 positions scored as high as the cipher, so the odds are be-

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low 1 in 20,000; counted exactly, the chance is at most about 2 in a billion [F4, F83]. Section 5 says what this does and does not measure. The other thirteen columns, 182 symbols in all, use exactly 13 distinct symbols, and those occur almost equally often: from 9 to 20 times each, statistically indistinguishable from uniform (chi-square p = 0.73) [F31].

The ciphertext as 196 two-digit symbols, 14 per row

75	62	82	85	91	62	91	64	81	64	91	74	85	84
64	74	74	82	84	83	81	63	81	81	74	74	82	62
64	75	83	82	84	91	75	74	65	83	75	75	75	**93**
63	65	65	81	63	81	75	85	75	75	64	62	82	**92**
85	74	63	82	75	74	83	81	65	81	84	85	64	85
64	85	85	63	82	72	62	83	62	81	81	72	81	64
63	75	82	81	64	83	63	82	85	81	63	63	63	**04**
74	81	91	91	84	63	85	84	65	64	85	65	62	**94**
62	62	85	91	85	91	74	91	72	75	64	65	75	**71**
65	83	62	64	74	81	82	84	62	82	64	91	81	**93**
65	62	64	84	84	91	83	85	74	91	81	65	72	74
83	83	85	82	83	64	62	72	62	65	62	83	75	**92**
72	63	82	82	72	72	83	82	85	84	75	82	81	83
72	84	62	82	83	75	81	64	75	74	85	81	62	**92**

Bold orange: the five rare symbols (04, 71, 92, 93, 94) — all in column 14 column 14

Figure 1. The challenge cipher laid out 14 symbols per row. All eight occurrences of the five rare symbols fall in the last column; the other 13 columns use 13 symbols.

That flatness is the cipher's central puzzle (Figure 2). Ordinary text is rarely this even: in 2,000 English passages of the same length, the thirteen commonest letters were this flat only once, and 182-letter passages of natural text use 18 to 25 different letters in English (median 22) and 14 to 25 in Latin (median 20), not 13 [F31, F81]. Tables that people ordinarily make rarely or never produce it: letters merged at random matched it in about 1 of 1,000 English passages (1 to 14 in 1,000 Latin ones, depending on the texts sampled), and two letters to a symbol in a fixed pattern never did [F31, F84]. What does reproduce it is a table built *on purpose* to balance the frequencies, in which each symbol stands for a group of letters chosen so that every symbol occurs about equally often: such tables match the cipher's flatness in about a third of trials for English (364 of 1,000). For Latin the rate depends on which texts are sampled: 345 to 374 of 1,000 for the ten-edition Latin corpus used, about one in six (167 to 182) for its six editions without English notes [F31, F80]. The English and Latin sources are listed in Appendix F. (A second purpose-built construction, a four-

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square with a key designed around the message, can also reproduce it; Section 7.) Beyond its flat counts the message carries no detectable order: rows, columns, halves and symbol-to-symbol dependence all look like a random shuffle (family-wise p = 0.91) [F32].

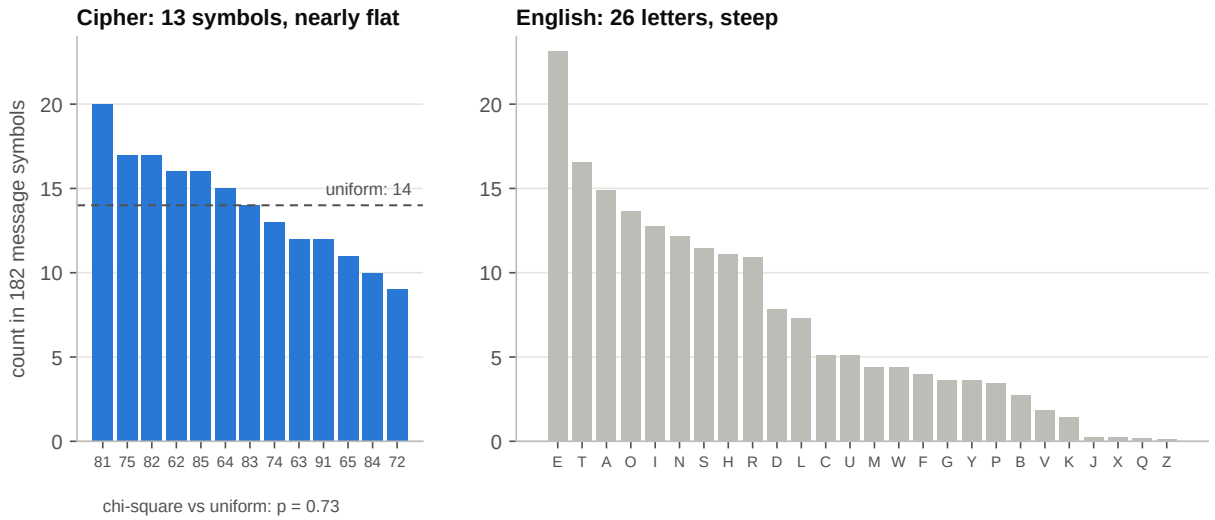


Figure 2. Left: the 13 message symbols occur almost equally often. Right: letters of English in a text of the same length, on the same scale. A one-to-one key keeps English's steep profile; the cipher does not have it.

3. Where the problem stood

The cipher has been discussed for decades. Wayne Barker described it for *Cryptologia* in 1978. Robert Matthews noted its flatness around 2003, and Erik van Eykelen first observed that its rare symbols fill one column [F4, C9].

Two substantial studies appeared in 2026. **Hendrick** showed that no one-to-one letter key, with any transposition, can produce the cipher's counts (the closest of 479,051 English passages needs 8 errors); that searches with measured power for columnar transposition, four-square with standard squares, coordinate shifts, homophonic keys, the book's own dummy rule and enciphering errors all come up empty; and that if a one-to-one key were used, the counts would point to Latin rather than English [C9]. Hendrick tested the cipher against a "no message" recipe and found it consistent (family-wise p = 0.28), with the caveat that the test barely detects a turning grille, and left the turning grille open: the search recovers planted grilles when the letter key is known (4 of 4) but not when grille and key are both unknown (0 of 4), and in Hendrick's tests under a Latin model it fails with keys only 65–69% right [C13, F34, O23]. **Gariazzo** rebuilt a solver scored by a statistical (quadgram) model of English and searched four cipher families — including a 14 × 14 Fleissner (turning) grille over an ordinary one-letter-per-cell square — with shuffled-cipher controls; none beat the controls [C12].

Two further 2026 preprints (doi:10.5281/zenodo.18639389 and doi:10.5281/zenodo.20565170) claim solutions, using a homophonic columnar transposition and a rail-fence-with-nulls construction respectively. They are noted here for completeness and have not been assessed.

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4. The hypothesis

What is observed. Three properties of the ciphertext itself:

- All eight occurrences of the rare symbols fall in column 14 of the 14 × 14 layout (odds below 1 in 20,000 on the row-length scan of Section 2) [F4, F83].
- The other 182 symbols use exactly 13 distinct symbols, almost equally often (chi-square against uniform, p = 0.73) [F31].
- No test we applied finds order among those 182 symbols: rows, columns, halves and symbol-to-symbol dependence all look like a random shuffle (family-wise p = 0.91) [F32].

What we infer, and how firmly. Each inference below rests on simulations of particular cipher families, and holds only for the families simulated:

- *The flatness points to a deliberate table.* In the simulations of this study, a one-to-one key on English was this flat once in 2,000 passages and letters merged at random about once in 1,000, while letters paired in a fixed pattern and the book's own sign system never were; a table built on purpose to balance the frequencies was, in about a third of English trials and in one in six to one in three Latin trials depending on the texts sampled [F31, F80, F65, R31, F84].
- *Such a table would need its order scrambled.* In the original 40 planted texts, a balanced table used without a transposition left detectable order (dependence between neighboring symbols beyond 400 shuffles, p < 0.05) in all 40 [R29]. A separately written check within the project, with new seeds and differently drawn practice texts, found it in 39 of 40, so the exact count depends on the texts sampled; both show strong detectability under the tested conditions [F82]. The cipher's own message shows no such dependence (p = 0.70).
- *A turning grille is one system that scrambles order thoroughly enough.* Of the systems Hendrick's order test examined, only a turning grille hides order this well [F34]; systems that test did not examine are not covered.
- *Column 14 is padding.* A simulation of the enciphering procedure supports reading column 14 as dummies added at the last step, with the message in the other 182 symbols [O16].

D'Agapeyeff's book shows he knew both ingredients: it mentions the grille (as Cardano's invention) and describes a sign system in which one sign serves for a pair of letters — g and j, f and v/y, s and z [F34, F42].

The hypothesis tested in depth in this report is therefore **a 14 × 14 turning grille over a balanced polyphonic table**. It is one construction consistent with the observations, not the only one: a four-square whose key is built on purpose around the message is also consistent with the frequency observations, and in some passages was not flagged by the order test of Section 7 [O34], and so is a cipher with no message at all [F66]. The message is written out in letters, each letter is replaced by the symbol of its group, the symbols are placed through the holes of a square card turned a quarter at a time (Figure 3), and column 14 is filled with dummies.

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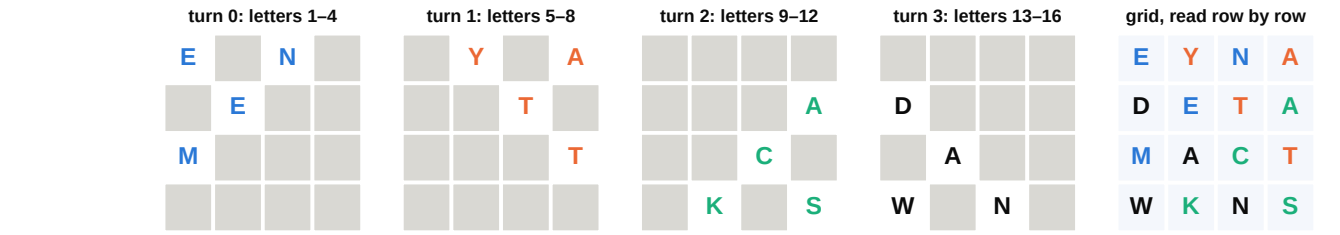


Figure 3. How a turning grille scrambles a message, shown on a 4 × 4 card. Each turn exposes four new cells; after four turns every cell holds one letter and the grid is read row by row.

A balanced table pools rare letters and leaves common ones alone, so that each of the 13 symbols occurs close to the ideal 1/13 (7.7%) of the time. The table below was computed from two million letters of English with the same balancing used in the experiments; it is an illustration, not the cipher's key:

Symbol	1	2	3	4	5	6	7	8	9	10	11	12	13
Letters	E	T	A	O	C·G·U	B·R	K·S	M·P·W	I	L·V·Y	H·Q·X	D·F·J·Z	N
Share	12.2%	9.1%	8.2%	7.8%	7.3%	7.1%	7.1%	7.0%	7.0%	7.0%	6.8%	6.7%	6.7%

E alone exceeds 1/13 and cannot be split, which is why even the best table is not perfectly flat. Reading such a cipher means choosing, for every symbol, which letter of its group was meant — with the true table and the true order, a standard decoding recovers 85–90% of the letters in practice ciphers [R29].

5. How we tested it

Practice ciphers. Every method was first run on ciphers we made ourselves in exactly the hypothesized form — a 14 × 14 grille, a balanced 13-symbol table, English text, dummies in column 14 — with the answer known. A method that cannot recover a practice cipher at this length tells us nothing when it fails on the real one [F68]. The plaintexts of the practice ciphers were drawn from a sample of English words weighted by frequency; because such text has no grammar, the identifiability measurements were repeated on natural prose (Section 6.5). Unless a sentence says otherwise, every success rate in Section 6 is a rate *in practice ciphers*, under the hypothesis; none is a statement about the real cipher's grille or table, which remain unknown.

Shuffled controls. A search can always find *something*; the question is whether it finds more in the real cipher than in the same symbols shuffled into random order. Comparisons use the same search, with the same effort, on shuffled copies.

The column-14 odds. The score of Section 2 rewards concentration *beyond chance*: with p columns, chance puts about one in p of the occurrences in any given column, so all eight in one column scores $1 - 1/p$ — 0.93 at row length 14, but only 0.5 at row length 2. Only eight in one column at a row length of 14 or more reaches the cipher's score, and the exact chance of that is at most about 2 in a billion; the published "below 1 in 20,000" is the floor that 20,000 random placements can show [F83]. A looser question — are all eight in one column at *some* row length from 2 to 20?

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— has a chance of about 0.7%, almost all of it from row length 2 (all eight in odd positions or all in even ones). That is a different question, and it is not the source of the odds in Section 2. The scan allows for choosing the row length; it does not allow for the choice to look at the rare symbols' columns in the first place, which was made after the alignment had been noticed (by van Eykelen) [F4, C9]. The wider search for structure, corrected for everything it examined, finds this one feature and no other [F16].

Judging a grille. A grille turned a quarter is the same grille, so a recovered grille was counted as correct if it matched the true one in any of its four orientations [F71].

Uncertainty. Rates are given with 95% Wilson intervals where the sample is small enough for the interval to matter. Results resting on fewer than ten practice ciphers per condition are preliminary demonstrations, not general success rates.

Recording claims. Every finding was recorded in the project's case file with its evidence. A claim proposed by an agent had to pass a programmatic claim gate — which refused claims that cited missing results, uncorrected p-values or statistics that shuffling cannot change — and be confirmed by a designated reviewer. Results produced by the developer were recorded directly, each with its evidence. Uncertain results are kept and labeled as such. A separate AI system, ChatGPT (OpenAI), reviewed several drafts of this report at the author's request; the checks it prompted are in Sections 6.5 and 7 and Appendix E, and its points on wording are taken up throughout. That is an AI review, not human peer review, and the case file itself has not been independently reviewed (Section 8). Section 10 describes the safeguards.

6. Results

6.1 With the true table supplied, practice-cipher grilles were recovered

If the polyphonic table is known, can the grille be found? Candidate grilles were scored by how English-like the resulting stream of *group* symbols is, using counts of symbol patterns from a 7-million-letter English corpus — 13 Project Gutenberg novels (Appendix F) — while the practice texts came from a different source, a sample of English words weighted by frequency. A grille counts as recovered when at least 45 of its 49 orbits are right, up to rotation. Scoring by six-symbol patterns recovered the grille on 6 of 6 practice ciphers (95% interval 61–100%): exactly — all 49 orbits — on 5, and with one orbit wrong on the sixth, where that near-miss scored slightly above the true grille. Five-symbol patterns also recovered 6 of 6 (47 to 49 orbits right, exactly on 3); four-symbol patterns, the usual choice in such searches, only 2 of 6 (10–70%) (Figure 4) [F78]. At five and six symbols the score never preferred a grille with fewer than 45 orbits right. Each search takes about two minutes on a laptop GPU. With six practice ciphers this is a demonstration that the method works, not a measured success rate.

In these practice ciphers, supplying the true table enabled successful grille recovery under five- and six-symbol scoring. Whether this generalizes to other keys, plaintexts and constructions remains untested. For the real cipher, the grille, the table and the construction itself all remain unconfirmed.

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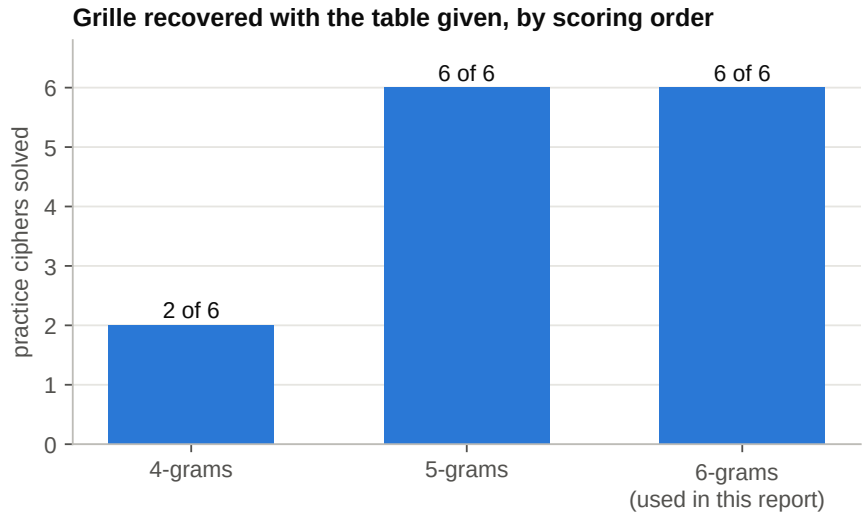


Figure 4. Practice ciphers: with the polyphonic table known, the grille is recovered (at least 45 of 49 orbits) on every practice cipher when candidate orders are scored by five- or six-symbol patterns; four-symbol scoring fails on four of six.

6.2 The table must be essentially exact

How close must the table be? The same search was given tables with some letters moved to the wrong group (keeping the groups balanced), four practice ciphers per condition. With the true table it found 4 of 4 grilles (51–100%); with 2 letters misplaced, 1 of 4; with 4 misplaced, 2 of 4; with 6, 8 or 12 misplaced, none of 12 (0–24%) (Figure 5) [F74]. With four trials, the 2- and 4-letter results cannot be told apart (intervals 5–70% and 15–85%); the clear contrast is between the true table and six or more misplaced letters. A table that is not nearly exact scores like a random one, so in these trials there is no gradual path toward the correct table. Hendrick found the same cliff for one-to-one keys [C13]; this result extends it, in a preliminary way, to polyphonic tables.

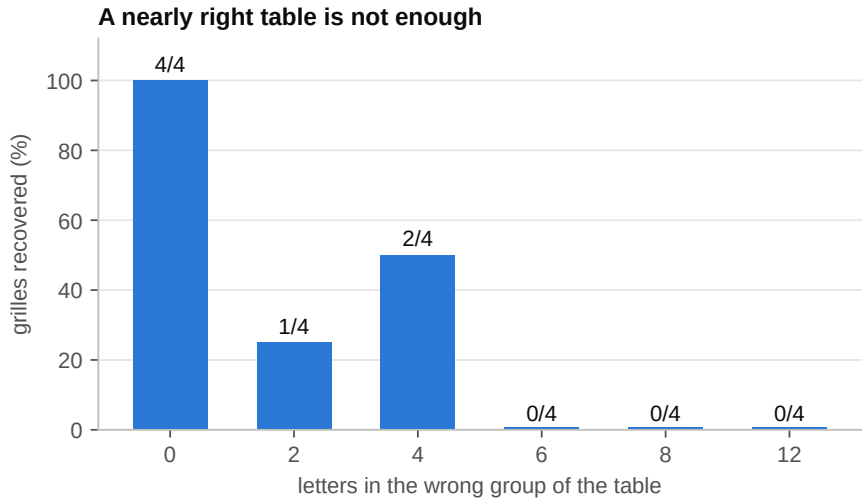


Figure 5. The same search given a table with some letters in the wrong group. From six misplaced letters on, it never finds the grille.

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6.3 Neither the cipher nor the book supplies the table

A natural next step is to repair a nearly-right table using the cipher itself. In one practice cipher whose grille had been recovered exactly, a search that changed the table to raise the score of the symbol stream moved the table *further* from the truth — from 2 wrong letters to 5. In a second practice cipher, whose grille was not recovered (19 of 49 orbits right), the table's error grew from 2 wrong letters to 8 [I47]. Section 6.4 measures this systematically.

Nor does the author's book help. Every table construction it describes or prints puts one letter in each cell of a 5 × 5 square. A message on 13 cells of such a square could use only 13 different letters, which the cipher's counts already exclude [F75].

6.4 The truth-start test: how well 182 symbols pin down the table

The decisive measurement asks: *if the grille were already solved,* does the text single out the true table under the score? For each of 100 practice ciphers per English model corpus (two corpora built from different books; here the two Wells novels belong to corpus A — Appendix F lists each corpus), the search is handed the true symbol order and **starts at the true table**. It then climbs by steepest ascent: every swap of two letters between groups is scored with the six-symbol score of Section 6.1, the best swap is taken, and this repeats until no swap raises the score. Two outcomes are possible:

- **Truth kept** — no swap raises the score, so the search ends at the true table: the true table is a local maximum of the score.
- **Higher-scoring table found** — the search moves away to a table that scores the text better.

In 3 of the more than 2,000 truth-start runs in this report the search moved to a table scoring higher by less than one part in a million — a numerical tie, where the passage cannot tell the two tables apart. These count as the truth *not* kept (and not as a higher-scoring table found); counting them as kept would change no figure by more than 2 points (corpus A at 364 symbols: 48% instead of 46%).

This is not blind recovery of an unknown key. The search is given both the true order and the true table, which is the most favorable case possible; a search for grille and table starting from nothing did worse (0 of 3 practice ciphers at every budget tried) [F68]. The table below and Figure 6 give how often the truth was kept [F76]:

Text length (symbols)	182	273	364	455	546	728	1,092
Corpus A	14%	31%	46%	65%	72%	84%	94%
Corpus B	8%	22%	36%	49%	54%	65%	74%
Both	11%	26.5%	41%	57%	63%	74.5%	84%

At the cipher's own length the search found a higher-scoring table in 178 of 200 practice ciphers (89%; 95% interval 84–93%), and kept the truth in 22 (11%; 7–16%). When letters were also allowed to move between groups, it found a higher-scoring table every time [F76]. The rate of keeping the truth rises steadily with length and passes one in two at roughly 420 symbols — more than twice the cipher's 182.

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What these numbers measure is narrow, and it matters. They are rates for this search and this score. A higher-scoring table shows that the true table is *not optimal under the tested score*; it does not establish which table is more probable under the true distribution of English. And because the search only looks at single swaps from the truth, 89% is a lower bound on how often a higher-scoring table exists under this score — not a measure of ambiguity in any information-theoretic sense. Section 6.5 tests how much the numbers depend on the procedure.

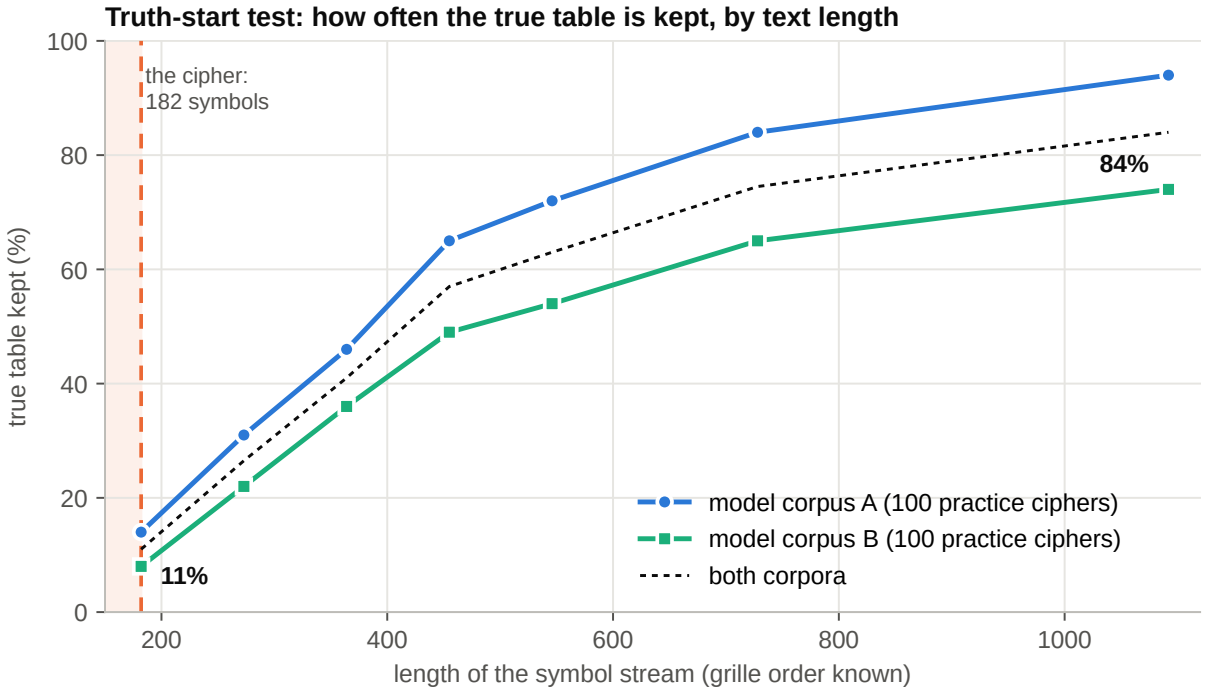


Figure 6. Truth-start test: how often the true table stays the best-scoring among its neighbors when the grille order is already known, by length of text, for two independent English models (100 practice ciphers each). At the cipher's 182 symbols it is about one in ten.

6.5 Robustness checks after review

The first review by ChatGPT asked three questions of Section 6.4: whether the results depend on the practice plaintexts, whether they depend on the language model and score, and whether the higher-scoring tables are genuine competing explanations or artifacts of the model. We reran the truth-start test with 50 practice ciphers for each of ten setups and four lengths (2,000 runs) [F77]:

- **Plaintext.** Natural prose — two novels by H. G. Wells, removed from every language model — instead of the word-frequency sample.
- **Corpus.** The two corpora of Section 6.4 (with the Wells novels taken out of corpus A) and both combined (6.6 million letters).
- **Score.** The six-symbol score used throughout, and a chain-rule score in which each symbol is predicted from the three, four or five before it.

Truth kept	182 symbols	273	364	546
Lowest of the ten setups	8%	10%	16%	28%

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Truth kept	182 symbols	273	364	546
Highest of the ten setups	26%	42%	44%	60%
All 500 runs at each length	15%	26%	29%	41%

Each setup's rate rests on 50 practice ciphers, so individual rates carry intervals of roughly ±10 points (for example 8%: 3–19%; 26%: 16–40%); the pooled 15% at 182 symbols has 12–18%. Appendix E gives every setup and length with its counts and interval, the data splits and the seeds.

No setup changes the conclusion. Natural prose did not make the truth easier to keep (12–18% at 182 symbols, against 20–26% for the word sample under the same models), and neither the combined corpus nor the chain-rule scores raised it (8–16% on prose at 182 symbols, against 12–18% for the original score).

What are the higher-scoring tables? Each table found at 182 symbols was scored two more ways, and the procedure matters for reading the result. The table was chosen by one search only: under one model, on one passage. It was then scored, unchanged, (a) on the *same passage* under the model of the *other, disjoint corpus*, with the same kind of score; and (b) on a *fresh passage* — 3,000 letters further on in the same source, enciphered with the same true table — under the *same model*. The fresh passage played no part in the search, and neither passage is in any model's corpus. Under the other corpus the table found still beat the true table in 69% of cases (171 of 248; 63–74%), so it is not an artifact of one corpus's sparse counts. On the fresh passage it beat the true table in only 19% (82 of 427; 16–23%). The higher-scoring tables are therefore fitted to the particular 182 symbols — passage-specific overfitting — rather than better descriptions of English. This explains the result of Section 6.4; it does not by itself establish non-identifiability in an information-theoretic sense.

Are scores comparable across tables? A later review by ChatGPT asked whether these comparisons are fair. Each candidate table is scored with a language model *rebuilt for that table*: the corpus is mapped through the candidate and its group patterns are counted. That is the right quantity in principle — it estimates how probable the symbol stream is if that table was used — but it could still favor some tables for reasons unrelated to the text, through the floor given to unseen patterns, smoothing, or the sizes of the groups. We checked three things [F79]:

- **Normalization.** The six-symbol model, with its floor for unseen patterns, does not sum to one (2.10 for corpus A′, 1.45 for corpus B), but its total varies by only 0.002 in log10 across the true table and its 305 single-swap neighbors.
- **Consistency.** Under a sufficiently accurate model of the true symbol distribution, the true table should outperform competing neighbors on sufficiently long held-out text. The finite-sample tests of this study assess whether that expectation holds for the models and neighbors examined here. On Wells prose held out of every model, no tested neighbor outscored the true table at 10,000 or 100,000 symbols — 0 of 9,150 comparisons over two corpora, three scores (six-symbol, six-symbol renormalized to sum to one, chain-rule) and five different true tables — against 1–3% of neighbors at 182 symbols and at most 1 in 1,525 at 1,092.
- **Renormalized rerun.** The truth-start test on prose with the renormalized six-symbol score kept the truth in exactly the same practice ciphers as the original score (200 of 200 runs: 9 and 6 of 50 at 182 symbols, 25 and 22 of 50 at 546).

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These checks support the interpretation that the higher-scoring alternatives arise primarily from short-passage effects rather than the tested normalization procedure. They do not exclude every possible model or scoring bias.

7. What it means

Shannon defined the *unicity distance* of a cipher system as the length of ciphertext beyond which only one key is expected to give a meaningful reading; below it, several keys fit equally well. Von zur Gathen (2022) estimated it at 80–152 symbols for the Zodiac-340 cipher, and argued that this supports the accepted solution. Shannon's quantity is defined by the true probabilities of the language. The present analysis is an empirical measurement under n-gram scores, the practical analog for the grille-plus-polyphonic system: in practice ciphers of the cipher's length, with the grille already solved, the true table was not the best under the score in three or more cases out of four, under every model and score we tested [F76, F77].

The hypothesis is therefore neither confirmed nor refuted by these methods: with them, the cipher is **too short to be decided from its own text**. A proposed solution of this kind would be one choice among several tables that score as well or better, and these scores could not show that it is D'Agapeyeff's. A much better model of the language might narrow the field; none we tried did. Two kinds of evidence would change the picture: the table itself, from a source outside the ciphertext (a notebook, a letter, a worked example), or a longer text enciphered the same way — on the present measurements at least twice as long, and under some of the models we tried three times or more.

Is it a question of computing time? Several searches, in this study and in earlier work, are recorded as having "no power" within a time budget — for the joint grille-and-table search, 1,200 seconds per phase [F68]. It is natural to read that as a shortage of computing power. What we can say is narrower:

- **Grille and table together.** The objective in every search here is an n-gram score of the symbol stream; the searches are simulated annealing over grille and table together (F68) and steepest ascent over table swaps (Section 6.4). For the tested scoring objectives, higher-scoring incorrect tables frequently exist even when the true symbol order and table are supplied — in 89% of practice ciphers [F76], and in 74% to 92% under every model and score of Section 6.5 [F77]. Consequently, additional optimization of the same objective cannot be assumed to improve true-table recovery. We have not established how success rates change with longer search budgets; the joint search found none of 3 practice grilles at every budget tried [F68], but that is not a budget study. A differently designed search or objective is not ruled out.
- **Any other transposition under a polyphonic table** (columnar, double): the same — the curve of Section 6.4 assumes the order is already known, and the true table is still usually not the best under the score.
- **Columnar or double transposition with an ordinary one-to-one key**: excluded by the symbol counts whatever the transposition [C9], and columnar was in any case searched with demonstrated power [R21, R27].
- **Four-square and two-square with random keys**: the randomly keyed constructions tested did not reproduce the observed 13-symbol usage in 10,000 practice ciphers [R35]. That excludes those sampled constructions under the tested plaintext (the English word sample), not every possible key or plaintext. In these systems each cipher symbol is set by the row of one plaintext letter and the column of another, so English spreads over nearly every symbol: 22 to 25 distinct symbols in 182 with 5 × 5 squares (20 to 25 even when the message is restricted to its 13 commonest letters) and 18 to 20 with 4 × 5 squares — none used 13 or fewer. The cipher's message uses exactly

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13.

A second construction that fits. The four-square exclusion holds only for keys chosen at random. An encipherer who knows the message can arrange the plaintext squares so that its frequent letters share a few rows and columns. A hill-climb doing exactly that — which, like the encipherer, uses the plaintext — reached 13 symbols in 13 of 100 passages of 182 letters (95% interval 8–21%) and 14 or 15 in most of the rest; all 13 were as flat as the cipher's counts (chi-square p from 0.85 to 1.0, against the cipher's 0.73). We then applied an order test — three mutual-information statistics (between the two symbols of each pair, in both phases, and the largest at lags 1 to 26), each against 2,000 shuffles of the same symbols, with a family-wise min-p correction, at the 5% level. Purpose-built keys were flagged less often than random keys: 12 of 100 (7–20%) against 52 of 100 (42–62%). Of the 13 passages that match the cipher's usage and flatness, 3 were flagged (passages 34, 77 and 95), leaving 10 of 100 that match the usage and flatness and were not flagged; the real message is not flagged (family-wise p = 0.999) [O34]. Not being flagged is not evidence that no order is present: the test examines three statistics only, and on the uncorrected pair statistics 7 of the 13 show dependence at $p < 0.05$.

This is a feasibility result. It shows that an intentionally constructed four-square *can* reproduce the cipher's 13 symbols and its flatness (13 of 100 passages) and, in 10 of 100, also go undetected by the order test we applied — the same kinds of observation that led to the grille hypothesis. It does not show that D'Agapeyeff used a four-square. The construction has not been searched: an ordinary four-square search cannot climb to the key [R32], and a key built this way has a structure that a new search would have to exploit. In this study's assessment — a judgment, not a measured probability — it is the most promising lead this study leaves open.

The alternative that the cipher carries no message remains open, as Hendrick concluded, and the replication of Hendrick's order test in this study is consistent with it (family-wise p = 0.29; 0.25 to 0.59 across 20 reruns of the same test) [F66].

8. Limitations

Sample sizes. The truth-start curve rests on 100 practice ciphers per corpus at each length and the checks of Section 6.5 on 50 per setup and length; the purpose-built four-square result on 100 passages. The grille-recovery and table-error results rest on 6 and 4 practice ciphers per condition, and the repair experiment of Section 6.3 on 1 (a second without the grille recovered) [F78, F74, I47]: they are preliminary demonstrations, with the wide intervals given in Section 6. The truth-start result (Sections 6.4 and 6.5) is the best supported.

Language model. Only English was tested. The two corpora agree on the shape of the curve but differ in level (14% against 8% at 182 symbols; 94% against 74% at 1,092), so the exact figures depend on the model. The checks of Section 6.5 — natural prose, a combined corpus, chain-rule scores — kept the truth at 182 symbols in 8% to 26% of practice ciphers, but they are all n-gram models; a stronger model of English (for example a neural one) was not tried.

Plaintext. The practice messages were drawn from a frequency-weighted sample of English words and, in Section 6.5, from held-out literary prose (two novels by H. G. Wells); the language models were built from novels. The results have not been tested on telegram-style messages, abbreviations, unusual vocabulary, deliberate misspellings or other kinds of plaintext that may be relevant to the historical cipher, and recovery and identifiability rates may differ for them. This does not affect the robustness findings of Section 6.5, which hold for the plaintexts tested.

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Search method and what it measures. The truth-start search climbs from the true table to any higher-scoring neighbor. Finding a higher-scoring table shows that the true table is not optimal under the tested scoring function; it does not establish which table is more probable under the true language distribution. Failing to find one does not show that the true table is the global best under the score. The 89% is therefore an empirical lower bound on how often a higher-scoring alternative exists under this score and procedure, and the rates of keeping the truth are rates for this procedure — neither is a probability that the table is identifiable by every possible method.

Review. The results have been checked within the project — against their evidence, by the claim gate, by fact-checking every number in this report, by reproducing the budget from the raw records, and by rerunning the headline experiments from a fresh copy of the repository (Section 10). ChatGPT (OpenAI) reviewed several drafts at the author's request, which led to Section 6.5, Appendix E, the four-square finding and more careful wording; that is an AI review, not human peer review or replication. Most case-file entries were recorded and confirmed by the same hand (the developer, or Turing for the team's results). Replication outside the project, which the repository is built to make easy, is the natural next step.

Score comparability. Section 6.5 provides empirical evidence that the scores are comparable across tables and consistent as the text grows, for the n-gram scores used; it does not cover other kinds of model.

Assumptions. The experiments assume a 14 × 14 turning grille, column 14 as dummies, exactly 13 balanced groups and English. A different layout, an unbalanced table or another language is not covered. The purpose-built four-square test (Section 7) used one simple hill-climb and 5 × 5 squares, and judged order by three dependence statistics.

9. Prior work, replications and new findings

Finding	Status
The counts are flat; the message has no detectable order (F31, F32)	Prior work — Matthews c. 2003; Hendrick 2026. Replicated in this study, with quantified rates.
No one-to-one key with any transposition fits the counts (C9)	Prior work — Hendrick 2026.
A turning grille is recovered in practice ciphers when the letter key is known (F71)	Prior work — Hendrick 2026; Gariazzo 2026. Replicated in this study.
The grille needs a nearly exact one-to-one key	Prior work — Hendrick 2026.
With the true **polyphonic** table supplied, six-symbol scoring recovered at least 45 of 49 grille orbits in six of six practice ciphers, five exactly (F78)	Extended in this study — Hendrick's key-given grille recovery, applied to polyphonic tables; a preliminary demonstration (95% interval 61–100%), not recovery of the historical grille.
A polyphonic table must be essentially exact (F74)	Extended in this study — Hendrick's exact-key result, applied to polyphonic tables; preliminary (4 practice ciphers per condition).

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Finding	Status
Under the n-gram scores tested, 182 symbols rarely single out the true polyphonic table, even with the grille known; measured curve to 1,092 symbols, robust across plaintexts, corpora and scores; the higher-scoring tables are fitted to the passage; scores comparable across tables (F76, F77, F79)	New finding, within the literature reviewed — a measurement under the tested scores; the study's main result.
A polyphonic table without transposition is excluded, under the tested conditions (R29)	New finding, within the literature reviewed — less certain: not found in the works checked, which may not cover it.
The book teaches no table that fits the hypothesis (F75)	New finding, within the literature reviewed — narrow and less certain.
Four-square and two-square with random keys excluded by symbol usage (R35)	New finding, within the literature reviewed — less certain: not in Hendrick or Gariazzo; not checked against the wider literature.
A purpose-built four-square can reproduce the cipher's symbol usage and flatness (13 of 100 passages; 10 of 100 also not flagged by the tested order statistic) (O34)	New finding, within the literature reviewed — a feasibility result and an open lead.

"Within the literature reviewed" means not found in Hendrick (2026) or Gariazzo (2026), read in full, nor in the abstracts returned by searches of OpenAlex, Semantic Scholar and arXiv on October 14, 2026. These novelty assessments are provisional: they rest on the literature examined for this report and do not establish priority over unpublished work, historical investigations or literature the search did not find. Specialist review may identify earlier equivalent methods or results.

10. How this research was done

This is an AI-assisted study, and the method is disclosed in full so that readers can weigh it.

The author. Nicolas Houghtaling (ORCID 0009-0001-3192-544X) conceived the study and directed it throughout, and is responsible for its conclusions. The author chose the D'Agapeyeff cipher and the question of whether it can be solved; designed and owns the PegasusDMZ research platform the work ran on, built with AI assistance; and made the decisions that shaped the research — insisting the team keep a record of established facts so that nothing was rediscovered (the origin of the case file), asking for a separate check of the team's first round, pressing for methods that had not been tried and for the weaknesses others had overlooked (the question that led to the flatness result of Section 2), approving the literature review that set this study alongside the work of others on the cipher — above all Hendrick's and Gariazzo's — aiming the final phase at the key itself (which led to the results of Section 6), stopping work that was not productive, and requiring that every claim be checked against the literature and stored with its evidence before anything is published, and, when AI reviews of the draft (ChatGPT) questioned the strength of its claims, holding publication until the checks they suggested had been run and the headline experiments had been reproduced from a fresh copy of the repository. Section 11.3 lists these decisions with their dates.

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The research platform. The work ran on *PegasusDMZ*, a self-hosted platform built by the author. It hosts a team of AI agents (language models given specialist roles such as statistician, historian and cryptanalyst) whose internet access passes through a governed proxy; Section 11.2 introduces each of them. The agents ran on DeepSeek, Kimi and Qwen models through DeepInfra. Software engineering, the search engines, the benchmark design and the final verification were done by Anthropic's Claude (Opus 5.5) working with the author as developer.

Safeguards against fooling ourselves. Language-model agents make confident mistakes, and several did during this work (an invented number in a consultation, a wrong statistic, a mislabeled claim of novelty). The platform therefore enforces: a *case file* in which every claim is recorded with its evidence, and an agent's claim stands only once a designated reviewer confirms it; a *claim gate* that refuses claims citing missing results, uncorrected p-values or failed controls; known-answer *practice ciphers* for every method before it touches the real cipher; *shuffled-cipher controls*; and a linked record in which a disputed claim automatically flags every claim that rests on it. The figures in this report are drawn directly from the result files, not retyped.

Review of the drafts. At the author's request, drafts of this report were reviewed by ChatGPT (OpenAI), an AI system separate from both the developer and the agent team, over several rounds. Its reviews prompted the robustness checks of Section 6.5, the score-comparability check, Appendix E, the internal audit and many narrower claims. Each suggestion was checked against the evidence before it was adopted, and a few were not adopted where the project's records did not support them. This is an AI review, not human peer review, and no one outside the project has yet replicated the results.

Cost. About US$68 of language-model usage for the agent team, a Claude subscription for the developer work, a ChatGPT Plus subscription for the ChatGPT reviews of the drafts, and a laptop GPU. Section 11 gives the full accounting.

Code and data. Every engine, experiment script, result file and figure behind this report is in the accompanying repository `dagapeyeff-cryptanalysis-1` (https://github.com/voodoomods/dagapeyeff-cryptanalysis-1), laid out so the scripts run unchanged; Appendix C lists them with their fingerprints.

Internal reproduction. Before publication the headline experiments were rerun from a fresh copy of the repository, following its README exactly — including rebuilding the language-model text, which reproduced its published fingerprint — inside a container with no network access, on the same laptop. Every rerun matched the published result run for run: the grille benchmark of Section 6.1 (18 of 18 runs), the truth-start test of Section 6.4 (the first 20 practice ciphers per corpus at 182 and 546 symbols, 80 of 80), the robustness checks of Section 6.5 (the first 10 practice ciphers of every setup at 182 and 546 symbols, 200 of 200), the purpose-built four-square search (the first 30 passages, 30 of 30) and the order test of Section 7 (identical summary). Beyond the outcomes, every regenerated record is identical to the published one in every field — scores included — except its elapsed time; the order-test file is byte-for-byte identical, and the other files' SHA-256 fingerprints differ only because they record run times. The scripts and the comparison are in `lab/cipher/round12/repro/`. This is an *internal reproduction*: it shows the repository reproduces its own results on the same hardware; it is not independent external replication. On October 16–17 every published experiment was then rerun at full size from a fresh clone of the release candidate, again with no network: of 4,959 published records compared, 4,956 are identical apart from elapsed time (the other 3 are search controls bounded by wall-clock time), and every number printed in this report was recomputed from the reruns; the record, with the one correction it led to (F81), is in `docs/validation-2026-10-16/`. Separately, Proof — the team's replica-

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tion reviewer, running on DeepSeek-V4-Pro — recomputed about 120 numbers in Sections 1, 6, 7 and Appendix E from the result files without seeing the developer's working. Its final verdict: every number matches the result files, apart from two rounded values (now shown exactly) and the overstated grille result also found while preparing the reproduction (corrected by F78). A first pass had also flagged two counts; they traced to an unstated convention — three runs in all ended in a numerical tie, which the report counts as the truth *not* kept — and the convention is now stated (Section 6.4). A follow-up internal audit — run within the project by the developer, with checking code written separately from the experiment code, and not a replication by an unaffiliated researcher — rebuilt all 100 four-square passages and their flatness tests, repeated the order test with fresh shuffles, re-derived the n-gram score, reran ten truth-start cases with its own search, and regenerated all 2,000 robustness runs to compare the greedy label matching recorded in `ident-full.json` with an exact assignment. Everything matched the published records; greedy and exact matching differ by one letter in 80 of the 2,000 runs, which changes no truth-kept or higher-scoring outcome and no figure in this report. It found one reporting issue — the four-square summary had merged the frequency result (13 of 100) with the order result (10 of 100 not flagged) — now separated (O34). The audit's scripts and outputs are in `lab/cipher/audit/`, its report in `docs/audit-2026-10-15/`. The audit and that adjudication are in `report/AUDIT-proof.md`.

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11. Budget, models and timeline

11.1 What it took

Every agent model call passes through the platform's proxy, which records the tokens and prices each call; the developer's usage comes from its session records. Figures cover the D'Agapeyeff study from October 8 to a snapshot taken on October 14, 2026; `evidence/accounting-2026-10-14.json` in the repository holds the aggregates and `tools/accounting.py` recomputes them from the raw records.

Resource	Calls / jobs	Input tokens	of which from cache	Output tokens	Cost
AI agent team (DeepInfra), on D'Agapeyeff tasks	10,570 calls	422.6 M	390.5 M (92%)	3.68 M	US$67.35

Resource	Calls / jobs	Input tokens	of which from cache	Output tokens	Cost
Planning chat with the chief-of-staff agent	30 calls	0.58 M	0.48 M (82%)	0.01 M	US$0.16
Developer: Claude Opus 5.5 (Claude Code)	1,234 calls	607.5 M	599.2 M (99%)	1.52 M	Claude Pro subscription — no per-token charge
Draft review: ChatGPT (OpenAI)	not recorded	—	—	—	ChatGPT Plus subscription — no per-token charge
Lab computer (laptop, NVIDIA RTX 5060)	89 jobs (54 on the GPU)	—	—	—	72 GPU-hours, 27.5 CPU-job hours; no charge

Most input tokens are re-sent context served from the providers' prompt caches, which is why the agents' 423 million input tokens cost about US$67. The developer's figure is the whole working session over the study and includes some work on the research platform itself done during it. The study built on tools first written for the team's earlier cipher work (an Enigma message and the Beale ciphers, from October 7), which is not counted here. The checks of October 15 (Sections 6.5 and 7) are not in the table either: they used no agent model calls and six lab jobs, about 3 GPU-hours and 0.7 CPU-job hours, plus about one GPU-hour for the reproduction run described in Section 10. The second review's numbers audit by Proof, the team's replication reviewer, used a small number of agent model calls (Section 10).

Agent spend by model (D'Agapeyeff tasks; each agent is introduced in Section 11.2):

Model	Role	Calls	Cost
DeepSeek-V3.2	the cipher specialists (Friedman, Welchman, Rejewski, Kahn, Kasiski, Painvin, Ventris and others)	8,608	US$48.99
DeepSeek-V4-Pro	review board (Fisher, Skeptic), chief of staff; the cipher room from October 14	1,287	US$9.93
Kimi-K2.6	Turing, chief cryptanalyst (to October 14)	616	US$8.20
Qwen3-235B-A22B	fallback when another model was busy	59	US$0.23

By day (UTC): Oct 9 US$1.43 · Oct 10 US$1.99 · Oct 11 US$39.83 · Oct 12 US$11.36 · Oct 13 US$7.84 · Oct 14 US$4.90.

11.2 The team

The agents are AI models (language models given a role, instructions and tools), **not people**. Most are named in tribute to figures from the history of codebreaking, statistics and scholarship; the names describe the role, and nothing in this report should be read as the work or opinion of the historical person, or as implying any association with or endorsement by them or their estates. Each ran on the model shown, at the cost shown (D'Agapeyeff tasks only).

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Turing — Chief cryptanalyst: plans the attack, divides the work, judges results

> Kimi-K2.6 (DeepSeek-V4-Pro from Oct 14) · 593 calls · US$8.32 · named after Alan Turing, Bletchley Park
>
> Led every round and split it into parallel angles; reviewed and recorded the team's findings (F65, F66, F68, R27, R32); drafted the body of this report

Friedman — Statistical cryptanalyst

> DeepSeek-V3.2 · 1,195 calls · US$7.31 · named after William F. Friedman, US Army cryptanalyst
>
> Statistics angles: what process gives 13 heavy and a few rare cells, what column 14 is, the book's dummy rules; replicated Hendrick's no-message test (F66)

Welchman — Search engineer: turns attacks into fast code, estimates run time first

> DeepSeek-V3.2 · 1,089 calls · US$6.60 · named after Gordon Welchman, Bletchley Park
>
> GPU searches: coordinate seriation, double transposition, the book's transposition passes

Rejewski — Mathematical cryptanalyst: structure and key space

> DeepSeek-V3.2 · 1,038 calls · US$6.40 · named after Marian Rejewski, who first broke Enigma
>
> Repeats, isomorphs and layouts; implemented the book's method; cataloged historical tables (F65)

Kahn — Historian: context, provenance, the author's methods

> DeepSeek-V3.2 (V4-Pro on Oct 14) · 1,085 calls · US$5.20 · named after David Kahn, historian of cryptology
>
> Reconstructed the book's methods from searches of its text; found its merged-letter sign system (F42); its retrieved passages settled F75

Kasiski — Pattern analyst: structure nobody asked about, corrected for everything looked at

> DeepSeek-V3.2 · 955 calls · US$5.08 · named after Friedrich Kasiski, 1863 method for repeated-key ciphers
>
> Pattern sweeps of the raw text (F16); what column 14 holds (F28)

Painvin — Compound-cipher specialist

> DeepSeek-V3.2 · 740 calls · US$4.21 · named after Georges Painvin, who broke the ADFGVX cipher
>
> Layered attacks: two-block models, alternating alphabets, the 14 × 14 grid on the GPU

Ventris — Decipherment linguist

> DeepSeek-V3.2 · 726 calls · US$4.08 · named after Michael Ventris, who deciphered Linear B
>
> Was the plaintext English at all (other languages, R28); readability scans of results; second-layer simulations (F65)

Batey — Error analyst: transcription and enciphering slips

> DeepSeek-V3.2 · 483 calls · US$3.28 · named after Mavis Batey, Bletchley Park
>
> Compared published transcriptions; searched for single slips; built no-message models

Navigator — Researcher: outside sources through the governed proxy

> DeepSeek-V3.2 · 515 calls · US$2.43
>
> Sourced and checked the exact transcription (F1); searched the 1939 book on the Internet Archive; found the 2026 preprints

Knox — Puzzle setter: thinks from the cipher author's side

> DeepSeek-V3.2 · 331 calls · US$2.11 · named after Dilly Knox, Bletchley Park
>
> What D'Agapeyeff meant readers to do; keyword phrases from the book (F29)

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Mercator — Author specialist: the author's life, books and map-making habits

> DeepSeek-V3.2 · 252 calls · US$1.69 · named after Gerardus Mercator, cartographer
>
> Turned D'Agapeyeff's other books and map-grid habits into testable keywords

Fisher — Review board: statistics referee

> DeepSeek-V4-Pro · 353 calls · US$3.09 · named after Ronald A. Fisher, statistician
>
> Refereed the statistics of round 4

Skeptic — Review board: red team, tries to break claims

> DeepSeek-V4-Pro · 234 calls · US$1.98
>
> Red-teamed round 4

Chief of staff — The platform's main assistant: the author's point of contact

> DeepSeek-V4-Pro · 512 calls · US$3.47
>
> Planning with the author; follow-up tests (e.g. the book's methods, decoding with given tables)

Ten other teammates made occasional calls on these tasks (US$2.09 together). The **developer** is not an agent of the platform: it is Anthropic's Claude (Opus 5.5), working with the author in Claude Code (Section 10).

11.3 Contributions

Nicolas Houghtaling (author): conceptualization, research direction and supervision, the research platform and compute, decisions on scope, method and publication, and review of the report.

Claude Opus 5.5 (developer): software and search engines, benchmarks and analyses, figures, and the writing and editing of this report under the author's direction.

OpenAI's ChatGPT (draft reviewer, through ChatGPT Plus): critical review of the drafts at the author's request — the questions behind the robustness checks, the score-comparability check, Appendix E, the internal audit and many narrower claims. It did not run experiments or write the report; each suggestion was checked against the evidence before it was adopted.

The agent team: supporting investigation, replications, exclusions and literature, as below.

Every finding in the case file records who entered it and who confirmed it. The results this study adds (Section 9) were produced by the developer — Claude Opus 5.5 — writing the search engines and running the benchmarks on the lab GPU; the agent team's work supplied replications, exclusions and the literature behind them.

Finding	Produced by	Model
The research questions behind the flatness result ("what weakness have others overlooked") and behind the final phase ("aim it at the key"); the case file of established facts	Author — Nicolas Houghtaling	—
Grille recovered with a polyphonic table given (F73, corrected in F78); exact-table requirement (F74); identifiability curve (F76); table repair moves away from the truth (I47)	Developer — engines and GPU benchmarks	Claude Opus 5.5

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Finding	Produced by	Model
The decision to test the review's points before publishing	Author — Nicolas Houghtaling	—
Robustness checks (F77); random-key four-square excluded (R35); purpose-built four-square matches the symbol usage and flatness (O34)	Developer — checks prompted by the ChatGPT reviews	Claude Opus 5.5
Flatness (F31), no order (F32), polyphonic tables need a transposition (R29), procedure simulation (O16), the book's tables (F75, from Kahn's book searches), novelty checks (C9, C12, C13)	Developer — analyses and literature checks	Claude Opus 5.5
Natural tables are rarely or never this flat (F65)	Rejewski and Ventris; recorded by Turing	DeepSeek-V3.2 (Turing: Kimi-K2.6)
Replication of Hendrick's no-message test (F66)	Friedman; recorded by Turing	DeepSeek-V3.2
Joint grille-and-table search has no power (F68); columnar + substitution excluded (R27); four-square basin too flat (R32)	The cipher team's GPU runs; recorded by Turing	DeepSeek-V3.2 / Kimi-K2.6

The agents also caught and corrected each other — and were corrected — through the review process described in Section 10; several of their errors are recorded in the fact-check log in the repository.

11.4 Timeline

Dates are the author's local dates. Evidence: the case file, the round briefings, the lab job records and the platform's version history.

Day	What happened
Thu Oct 8	**The author chooses the D'Agapeyeff cipher** and directs the team to attack it from several angles; the transcription is sourced and checked.
Fri Oct 9	Round 1: the agent team's first attempts.
Sat Oct 10	**At the author's request**, a review of round 1 by the developer (Claude), separate from the agent team, finds methodological errors (the book's example square assumed as the key; shuffles of raw digits used as controls). New tools with built-in shuffled controls and known-answer self-checks; rounds 2–4. The author also asks that the team keep its established facts — the case file follows.
Sun Oct 11	The heaviest day (about 6,000 agent calls, US$40). The case file is created and the transcription confirmed; at the author's request a pattern specialist (Kasiski) joins; column 14 holds every rare symbol; columnar and substitution families excluded on the GPU; the author presses for untried methods and overlooked weaknesses, and the flatness (F31) and missing order (F32) follow; the author approves the literature check, which finds Hendrick's paper of October 8; polyphonic tables shown to need a transposition (R29); the turning-grille engine and polyphonic decoder are built. Rounds 5–8.
Mon Oct 12	Round 9. The book's own merged-letter sign system is found and excluded (F42, R31); the joint grille-and-table search is shown to have no power at this length (F68).

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Day	What happened
Tue Oct 13	Team results recorded: natural tables rarely or never this flat (F65), Hendrick's no-message test replicated (F66), four-square basin (R32). Then **the author aims the final phase at the key**; grille recovered with the key given (F71); the author stops an unproductive agent round and refocuses the work on the table; six-symbol scoring recovers the grille with the table given (F73); the exact-table requirement (F74); the book's tables (F75); table repair fails on practice ciphers (I47); the author asks whether the results are unpublished — full-text novelty check against Hendrick and Gariazzo (C12, C13) — and orders the evidence preserved; the identifiability study starts overnight on the GPU.
Wed Oct 14	The identifiability curve completes (F76). The author commissions this report and the repository; both are drafted, fact-checked and assembled.
Thu Oct 15	A review of the draft by ChatGPT (OpenAI), requested by the author, questions the strength of three claims. **The author holds publication** and has all four suggested checks run: natural prose, other corpora and scores, the nature of the higher-scoring tables (F77), and purpose-built four-square keys with an order test (R35, O34). A second review of a later draft asks for narrower wording, intervals, the full robustness breakdown (Appendix E) and a reproducibility audit; the headline experiments are rerun from a fresh copy of the repository. A third review asks whether scores are comparable across tables; the check passes (F79). An internal audit separates the four-square frequency and order counts (O34).

Acknowledgements

This study stands on the work of everyone who has taken the D'Agapeyeff challenge seriously. We thank Wayne Barker, whose 1978 *Cryptologia* article set out the problem; Robert Matthews, who noticed the flatness of the counts; Erik van Eykelen, who first saw that the rare symbols fill one column and proposed a construction with no message; Chase Hendrick, whose 2026 paper turned decades of conjecture into measured exclusions and a clear list of what remained open; Ruben Gariazzo, whose 2026 report and released code showed how carefully a solver must be checked against chance; the transcribers and archivists who preserved the cipher and D'Agapeyeff's book; Project Gutenberg and its volunteers, whose editions of the novels and Latin texts listed in Appendix F made the language models possible; and Robyn Speer's wordfreq, whose English data draws on SUBTLEX (freely available word lists by Marc Brysbaert and colleagues), Google Books Ngrams (http://books.google.com/ngrams), OpenSubtitles and other sources. We also credit ChatGPT (OpenAI), which reviewed drafts of this report at the author's request and whose questions led to Section 6.5, Appendix E, the four-square finding of Section 7, the internal audit and more careful claims throughout. Where this report goes further, it does so on their foundations.

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Glossary

Term	Meaning
Cell / symbol	One of the two-digit numbers in the cipher, read as a row digit (6–9, and once 0) and a column digit (1–5) of a coordinate grid.
Polyphonic table	A key in which one symbol stands for *several* letters (the reverse of a homophonic key, where one letter has several symbols).
Turning grille	A square card with holes. The message is written through the holes, the card turned a quarter, and so on four times; the filled grid is then read row by row.
Orbit	A set of four cells that a turning grille cycles through as it turns; a valid grille has exactly one hole in each orbit.
Practice cipher (plant)	A cipher we made ourselves with a known key, used to measure whether a method can find the answer.
Shuffled control	The same search run on the cipher's symbols in random order, to see what "finding something" looks like by chance.
n-gram score	How likely a text is, judged by how often each run of *n* consecutive symbols occurs in a large sample of English.
Chain-rule score	A likelihood that predicts each symbol from the few before it and multiplies those predictions, as opposed to scoring each run of symbols on its own.

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Term	Meaning
Identifiable	The data single out one key as the best-scoring under a given model and score; when not identifiable in this sense, wrong keys score the text as well or better.
Unicity distance	The length of ciphertext beyond which only one key is expected to give a sensible reading (Shannon 1949).

Appendix A. Techniques built or adapted in this study

These are the methods this study built or adapted. None is claimed as a new algorithm: they were not found in the D'Agapeyeff literature checked (Section 9); they were not checked against the general cryptanalysis literature, and some are adaptations of standard tools. Paths are relative to the repository root.

A.1 Group-symbol n-gram scoring. To search a transposition when each symbol stands for a group of letters, we score candidate orders with an n-gram model over the 13 *group* symbols, built by passing a large English corpus through the table. No letter has to be chosen for any symbol, so the transposition can be searched separately from the decoding. Six-symbol patterns recover the grille on 6 of 6 practice ciphers (exactly on 5) where the usual four-symbol patterns recover 2 [F78]. Code: `lab/cipher/grille/grille_engine.py` (`group_ngrams`, `group_ngrams_ids`, `anneal(…, n, base)`).

A.2 Rotation-invariant judging of grille recovery. A turning grille turned a quarter is the same grille. A search may return the right grille in any of its four orientations, so recovered grilles are compared with the planted one in all four (`match_up_to_rotation`) [F71].

A.3 GPU-batched annealing over grille orbits. A 14 × 14 grille with a dummy column is 49 orbits of four cells, each taking one of four turns. Hundreds of annealing chains (512 in the runs reported here) move orbit turns in parallel on one GPU, with a fixed step budget so results do not depend on machine load (`anneal`). This is an implementation of standard simulated annealing, not a new algorithm.

A.4 Balance-preserving table perturbation. To measure how exact a table must be, letters are swapped between groups so that every group keeps its size, then the grille search is rerun with the perturbed table [F74]. Code: `lab/cipher/grille/bench_tolerance.py`.

A.5 Likelihood repair of a table with the order fixed. Given a symbol order, every swap of two letters between groups (and every move of a letter to another group) is scored by the n-gram likelihood of the symbol stream, rebuilding the group model on the GPU for each candidate; the best change is taken until none improves [I47]. Code: `repair_table` in `grille_engine.py`; `lab/cipher/round11/table_test.py`.

A.6 Truth-start test. Starting the repair search *at the true table* asks whether any nearby table scores the text better. Each time one does, the true table is shown not to be optimal under the score — so the rate is a lower bound on how often a higher-scoring alternative exists under that score, not a measure of ambiguity under the true language. Repeating this over text lengths and two independent language models gives the curve of Section 6.4 [F76]. Code: `lab/cipher/round11/identifiability.py`.

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A.7 Neighbor-pair key signal (exploratory). Under a turning grille, cells that are horizontally adjacent in the grid are often consecutive letters of the message (14–37% of message-column pairs in the practice ciphers of this study), so a likelihood built from neighboring pairs can rank keys without knowing the grille. A simple hill-climb using it beat frequency-only key accuracy in 4 of 6 practice ciphers, too weakly to help [O32]. Exploratory: the code was run interactively and is not preserved; nothing in the report relies on it.

A.8 Symbol-usage test for digraphic systems. Before searching a system, ask whether it can produce the cipher's symbol usage at all: encipher English with random keys and count the distinct symbols in 182-symbol windows. Four-square and two-square with random keys never come close to the cipher's 13 (0 of 10,000 practice ciphers) [R35]. Code: `lab/cipher/round11/micro_digraph_cells.py`, `micro_digraph_cells_4x5.py`.

A.9 Safeguards in the research process. A claim gate refusing claims that cite missing results, uncorrected p-values or statistics that shuffling cannot change; permutation tests that flag such invariant statistics and report a family-wise p-value; and a linked case file in which disputing a claim flags every claim resting on it. These are part of the PegasusDMZ platform (Section 10) and are not included in the repository.

A.10 Calibration of better-scoring tables. To tell a genuine competing table from one that merely fits the passage, every table that beats the true one is scored again under an independent language model and on a fresh passage enciphered with the true table [F77]. Code: `lab/cipher/round12/review_checks.py` (`ident`).

A.11 Purpose-built key search and pair-dependence test. For a given passage, a hill-climb rearranges the plaintext squares of a four-square to use as few cipher symbols as possible, the two cipher squares are aligned to make the counts as flat as possible, and the result is tested for order with mutual information between the two symbols of each pair (both phases) and at lags 1 to 26, against shuffles with a family-wise correction [O34]. Code: `lab/cipher/round12/review_checks.py` (`fsq`), `fsq_order.py`.

A.12 Score-comparability check. For a scoring model rebuilt per candidate table: measure the model's total mass across the true table and its neighbors, score the true table against every single-swap neighbor on held-out text of growing length (under a sufficiently accurate model, no neighbor should stay ahead on long text), and rerun the truth-start test with a renormalized score [F79]. Code: `lab/cipher/round12/comparability.py`.

Appendix B. Approaches that did not work

Approach	Outcome	Entry
Columnar transposition (width 14) + one unknown substitution, with the 04 kept or removed	Practice ciphers recovered; the real cipher scored below its shuffled control	R27, R21
Four-square / two-square with random keys	Excluded by symbol usage: they spread English over 18–25 distinct symbols, the message uses 13. A purpose-built key is not excluded (O34), but an ordinary four-square search cannot climb to the key (a key about 12 swaps from the truth)	R35, R32

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Approach	Outcome	Entry
Grille and table searched together, nothing known	0 of 3 practice ciphers at every budget tried. For the tested objectives the true table is usually not the best-scoring one even when supplied (F76), so more optimization of the same objective cannot be assumed to help; longer budgets were not studied	F68, F76
Dictionary keyword squares ranked by how well cell counts fit English (148,398 squares)	None beats shuffled cell labels	R26
Naturally made polyphonic tables (merges, letter pairs, keyword tables)	Rarely as flat as the cipher (random merges about 1 in 1,000 English passages) or never (letter pairs, keyword tables); deliberately balanced tables are, in about a third of English trials	F65, F84
Neighbor-pair key recovery (A.7)	Real but weak signal	O32

Appendix C. Evidence

The key results of this report come from the files below; other figures come from case-file entries (Appendix D). The SHA-256 fingerprint (first 16 hex digits shown; `MANIFEST.sha256` in the repository has them in full) identifies each file exactly, so a reader can confirm that a file is the one the report used. Paths are in the `dagapeyeff-cryptanalysis-1` repository (https://github.com/voodoomods/dagapeyeff-cryptanalysis-1).

File in the repository	SHA-256 (prefix)
`evidence/case/dagapeyeff.json`	`9d7de561b6dd600e…`
`lab/cipher/grille/bench-2.json`	`d5bdb4eff4649c20…`
`lab/cipher/grille/bench2.py`	`47a27dd4e2ececf6…`
`lab/cipher/grille/bench-ngram.json`	`3d07abc78366047a…`
`lab/cipher/grille/bench_ngram.py`	`2b7912e681a1bd4a…`
`lab/cipher/grille/bench-ngram.json`	`3d07abc78366047a…`
`lab/cipher/grille/bench_ngram.py`	`2b7912e681a1bd4a…`
`lab/cipher/grille/bench-tolerance.json`	`00dbb2cff1eb8fe6…`
`lab/cipher/grille/bench_tolerance.py`	`8e4010738bb4414f…`
`lab/cipher/round11/validate.json`	`655dfc74cf267003…`
`lab/cipher/round11/identifiability-A.json`	`17a03e2ea96ca47b…`
`lab/cipher/round11/identifiability-B.json`	`4893babc18d44d12…`
`lab/cipher/round11/identifiability-A-big.json`	`747281bcd0f2b4f5…`
`lab/cipher/round11/identifiability-B-big.json`	`753cd63d3e40b33e…`

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File in the repository	SHA-256 (prefix)
`lab/cipher/round11/table_test.py`	`f5fdc2b1f1a669d2…`
`lab/cipher/round11/identifiability.py`	`37313841573108c5…`
not redistributed: extracts from D'Agapeyeff's 1939 book (archive.org full-text search) — not redistributed	`745eeffda23d5847…`
`lab/cipher/round11/micro-digraph-cells.json`	`ea0c8709e669b75a…`
`lab/cipher/round11/micro-digraph-cells-4x5.json`	`ab9aa0c03425536d…`
`lab/cipher/round11/micro_digraph_cells.py`	`c2676428e4a0a57e…`
`lab/cipher/round11/micro_digraph_cells_4x5.py`	`df6b4cd99899ed39…`
`lab/cipher/round11/micro-digraph-cells.json`	`ea0c8709e669b75a…`
`lab/cipher/round11/micro-digraph-cells-4x5.json`	`ab9aa0c03425536d…`
`lab/cipher/round11/micro_digraph_cells.py`	`c2676428e4a0a57e…`
`lab/cipher/round11/micro_digraph_cells_4x5.py`	`df6b4cd99899ed39…`
`lab/cipher/round12/ident-full.json`	`c3f786de75a98f8b…`
`lab/cipher/round12/ident-small.json`	`ae575e5c38975225…`
`lab/cipher/round12/review_checks.py`	`1d68304edeab7660…`
`lab/cipher/round12/comparability.json`	`48656c6f94b25d5a…`
`lab/cipher/round12/comparability.py`	`c8a4d82154715772…`
`lab/cipher/provenance/latin_flatness.json`	`94e69f512218c683…`
`lab/cipher/provenance/latin_flatness.py`	`ff81f367fa031c97…`
`lab/cipher/round12/fsq-full.json`	`a58d07779d90c6c7…`
`lab/cipher/round12/fsq-order.json`	`ca1074a493f8a197…`
`lab/cipher/round12/review_checks.py`	`1d68304edeab7660…`
`lab/cipher/round12/fsq_order.py`	`a00d85ffdfa843c0…`
`lab/cipher/round12/fsq-full.json`	`a58d07779d90c6c7…`
`lab/cipher/round12/fsq-small.json`	`eee3b3fef3975593…`
`lab/cipher/round12/fsq-order.json`	`ca1074a493f8a197…`
`lab/cipher/round12/review_checks.py`	`1d68304edeab7660…`
`lab/cipher/round12/fsq_order.py`	`a00d85ffdfa843c0…`
`evidence/jobs/lab_jobs.json`	`78a4c064c7d492f1…`

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File in the repository	SHA-256 (prefix)
`lab/cipher/grille/grille_engine.py`	`54f3503ff8444788…`
`lab/cipher/grille/poly_engine.py`	`fceb05c6fd6d86dc…`
`lab/cipher/grille/digraph_engine.py`	`8b459be99f022ff4…`
`docs/FINDINGS.md`	`d64868b4d14100aa…`
`docs/literature/NOVELTY-CHECK-2026-10-14.md`	`cd2145d704b4c963…`
not redistributed: literature search results incl. 31 third-party abstracts — not redistributed; titles, authors, years and DOIs are in docs/literature/novelty-search-2026-10-14.metadata.json	`8af129a6a7ea2d88…`
`docs/BRIEFING-round10.md`	`c3fa41b13af1d52d…`
`docs/literature/gariazzo2026/Report_Agapeyeff.pdf`	`f38efcfa610a90ec…`
not redistributed: Hendrick 2026 manuscript — all rights reserved by its author; cite doi:10.5281/zenodo.23241303	`e86315c2526beda8…`

Appendix D. Case-file codes

Every bracketed code in this report, such as [F78], is an entry in the project's case file — the running record in which each result is written down with its evidence. The letter gives its kind: **F** an established fact, **R** a system or idea ruled out, **I** a result that is suggestive but not conclusive, **C** a convention — chiefly prior work and how we credit it — and **O** an open question or lead. All entries below except F73 and R34 (retired) have the status *confirmed*, given by one of the case's designated confirmers (Turing, the review board or the developer) — in most cases the same hand that recorded the entry; O33 stays confirmed as a record, but its wording is superseded by O34. None has yet been reviewed outside the project. The full text of each, with its evidence and review history, is in `evidence/case/dagapeyeff.json` in the repository; where a result file is listed, it is in the repository too.

Code	What it says	Result file
C9	Prior work: the flatness of the counts (Matthews), and the proof that no one-to-one key with any transposition fits them (Hendrick 2026), were published first.	—
C12	Prior work: Gariazzo (2026) already ran a turning-grille search on this cipher with shuffled controls, so the key-given grille search of this study (F71) is a replication.	—
C13	Precedent: Hendrick (2026) measured how exact a one-to-one key must be for the grille search; neither Hendrick nor Gariazzo treats polyphonic tables, unicity distance or whether the table is determined.	—

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Code	What it says	Result file
F1	The transcription: 395 digits (78 groups of five plus “92000”), identical in every standard source and in the 1974 reprint.	`data/ciphertext.txt`
F4	With the 04 counted the text is 196 two-digit symbols, 14 × 14; all eight occurrences of the rare symbols sit in column 14 (odds below 1 in 20,000 on the row-length scan: 0 of 20,000 random placements scored as high; see F83).	`lab/cipher/round11/real_cells.json`
F16	A sweep for any structure in the original text — layouts, periods, repeats, neighbor rules, corrected for everything examined — finds exactly one: column 14.	—
F28	Column 14 holds the eight rare symbols and six ordinary ones.	`lab/cipher/round11/real_cells.json`
F29	A list of 308 period keywords and phrases (1930s context and the book’s own wording) was compiled for keyword tests.	—
F31	The 13 message columns use 13 symbols almost equally often (p = 0.73). Natural text and naturally made tables are rarely this flat (about 1 in 1,000 English windows; paired-letter tables never; see F84); deliberately balanced tables are, in about a third of English trials.	—
F32	Apart from its flat counts the message shows no order: rows, columns, halves and symbol-to-symbol dependence all look shuffled (family-wise p = 0.91).	—
F34	The book mentions the grille (as Cardano’s invention); Hendrick’s order test leaves only “no message” or a system that destroys order as completely as a turning grille.	—
F42	The book describes a sign system in which one sign serves for a pair of letters (g/j, f/v-y, s/z) and ch, sh, th have signs of their own.	—
F65	Historical and naturally made 13-symbol tables match the cipher’s flatness rarely or never (0 to near 0%); deliberately balanced ones match far more often.	—
F66	Hendrick’s 30-statistic “no message” test, replicated within this study by the team’s statistics specialist (not an independent replication): family-wise p = 0.29 (his: 0.28); 20 reruns of the same unseeded test give 0.25–0.59.	`docs/validation-2026-10-16/`

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Code	What it says	Result file
F68	Searching for grille and table together, with nothing known, has no power at this length: 0 of 3 practice ciphers at every budget tried.	—
F71	With the letter key given, the grille search recovers every planted grille exactly (3 of 3), counting the four orientations as one grille.	`lab/cipher/grille/bench-2.json`
F73	*Retired, superseded by F78.* Stated that six-symbol scoring found all 49 orbits on 6 of 6 practice ciphers; one was 48 of 49.	`lab/cipher/grille/bench-ngram.json`
F74	The table must be essentially exact: grille found 4/4 with the true table, 1/4 with 2 letters wrong, 2/4 with 4, never with 6 or more (0/12).	`lab/cipher/grille/bench-tolerance.json`
F75	Every table construction the book teaches puts one letter in each cell, so none fits the hypothesis.	(book extracts — fingerprint only)
F76	Truth-start test: with the grille order known, the true table stays the best-scoring among its neighbors in only 11% of 200 practice ciphers at 182 symbols, rising to 84% at 1,092 (six-symbol score).	`lab/cipher/round11/identifiability-A-big.json`, `-B-big.json`
F77	The F76 result holds under natural prose, three corpora and four scores (true table kept in 8–26% at 182 symbols); the higher-scoring tables fit the passage, not English (they win on a fresh passage only 19% of the time).	`lab/cipher/round12/ident-full.json`
F79	Scores are comparable across candidate tables: the model's total mass varies by 0.002 in log10 across tables; on 10,000 and 100,000 held-out symbols no tested neighbor outscores the true table (0 of 9,150); a renormalized score keeps the truth in the same 200 of 200 runs.	`lab/cipher/round12/comparability.json`
F80	Qualifies F31's Latin figure: with the committed tool a balanced table matches the cipher's flatness in 345–374 of 1,000 windows of the ten-edition Latin corpus, but 167–182 when windows come only from its six editions without English notes; the Latin rate depends on the texts sampled.	`lab/cipher/provenance/latin_flatness.json`
F81	Corrects F31's letter range: 182-letter windows use 18–25 different letters in English (median 22) and 14–25 in Latin (median 20), not "17 to 22"; still never 13.	`docs/validation-2026-10-16/`
F82	Clarifies R29: 40 of 40 is the result for the original planted texts and seeds (rerun identical). A separately written check with new seeds and differently drawn texts found order in 39 of 40. The count is not seed-invariant; the conclusion is unchanged.	`docs/validation-2026-10-17/`

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Code	What it says	Result file
F83	Clarifies F4: the statistic is, over row lengths 2–20, the largest share of the eight rare occurrences in one column less 1/p. “Below 1 in 20,000” is the floor of 20,000 random placements (none as high); the exact chance is at most 1.8 × 10□⁹. All eight in one column at *any* row length is a different event (about 0.7%, mostly row length 2).	`validation/f4_events.py`
F84	Qualifies F31 and F65’s “never”: in the standalone flatness runs, letters merged at random matched the cipher’s flatness in 1 of 1,000 English windows (2 of 2,000) and 1 to 14 of 1,000 Latin ones; tables with two letters per symbol in a fixed pattern matched in none. Rare, not impossible; balanced tables remain far more frequent.	`docs/validation-2026-10-16/flatness/`
F78	With the polyphonic table given, six-symbol scoring recovers the grille (at least 45 of 49 orbits) on 6 of 6 practice ciphers — exactly on 5, 48 of 49 on the sixth; five-symbol on 6 of 6; four-symbol on 2 of 6. Corrects F73.	`lab/cipher/grille/bench-ngram.json`
I47	Repairing a nearly right table by likelihood moved it further from the truth: with the grille already correct, 2 wrong letters became 5 (1 practice cipher); in a second, whose grille was not recovered, 2 became 8. Strengthened by F76.	`lab/cipher/round11/validate.json`
O16	A simulation of 304 enciphering procedures: column 14 is best explained as padding added at the last step, with the message in the other 182 symbols.	—
O23	Hendrick’s open problems: the turning grille (no search finds its key) and keyed four-square and two-square remain open.	—
O32	Under a turning grille, neighboring cells are often consecutive letters — a key signal that does not need the grille, but a weak one (4 of 6 practice ciphers improved).	—
O33	*Wording superseded by O34* (numbers unchanged): stated that a purpose-built four-square fits the cipher’s symbols, flatness and “mostly” its lack of order, without separating the 13 of 100 (frequency) from the 10 of 100 (frequency and order).	`lab/cipher/round12/fsq-full.json`, `fsq-order.json`
O34	A four-square whose key is built on purpose around the message matches the cipher’s 13 symbols and flatness in 13 of 100 passages; 10 of those 13 were not flagged by the tested family-wise order statistic at the 5% level; not yet searched.	`lab/cipher/round12/fsq-full.json`, `fsq-order.json`

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Code	What it says	Result file
R21	One columnar transposition of width 14 plus one substitution, with the 04 removed: practice ciphers solved, the real cipher scores below its shuffled control.	—
R26	No dictionary keyword square (148,398 tested) fits the cell counts better than shuffled labels.	—
R27	The same columnar-plus-substitution system with the 04 kept as a symbol (an aligned 14 × 14 grid): excluded in the same way.	—
R28	French, Italian, Spanish, Dutch and Latin, each with its own keyword squares, fit no better than chance.	—
R29	A polyphonic table used without a transposition left order in the text in all 40 of the original practice texts (see F82 for a check with new seeds); the cipher has none, so if such a table was used its order was scrambled.	`validation/original/r29_original.py`
R31	English written in the book's own sign system never gives 13 flat symbols (0 of 400 passages).	—
R32	Four-square and two-square searched by single swaps: the landscape is too flat to climb.	—
R34	*Retired, superseded by R35 and O33.* Keyed four-square and two-square cannot produce the cipher's symbol usage — true for random keys only.	`lab/cipher/round11/micro-digraph-cells.json`, `-4x5.json`
R35	Four-square and two-square with random keys cannot produce the cipher's symbol usage: they spread English over 18–25 distinct symbols in 182 (0 of 10,000 practice ciphers at 13 or fewer); the message uses 13.	`lab/cipher/round11/micro-digraph-cells.json`, `micro-digraph-cells-4x5.json`

Appendix E. Robustness breakdown

Every setup and length of Section 6.5, from `lab/cipher/round12/ident-full.json` [F77]. Each row is 50 practice ciphers (plants 0–49). *Truth kept*: the search, started at the true table, ended there — no swap of two letters between groups raises the score, so the true table is a local maximum of that score. *Higher-scoring found*: the search moved to a different table with a higher score. Of those, *other corpus*: the table found also beats the true table under the other corpus's model on the same passage (setups with a disjoint second corpus only); *fresh passage*: it also beats the true table under the same model on a passage 3,000 letters further on, which played no part in the search. Intervals are 95% Wilson intervals. One run (Wells prose, A′ + B, chain-rule 3 before, 273 symbols) ended in a numerical tie and counts in neither column (§6.4). The result file also records how many letters the table found differs from the truth, using a

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greedy match of group labels; an exact assignment gives identical truth-kept and higher-scoring outcomes in all 2,000 runs and differs from the greedy count by one letter in 80 (internal audit, `lab/cipher/audit/label_full.json`). Those letter counts are not used in this report.

Plaintext	Corpus (letters)	Score	Length	Truth kept	95% CI	Higher-scoring found	other corpus	fresh passage
word sample	A′ (5 books, 1.95 M)	6-symbol (as §6.4)	182	13 of 50 (26%)	16–40%	37	23 of 37	1 of 37
word sample	A′ (5 books, 1.95 M)	6-symbol (as §6.4)	273	21 of 50 (42%)	29–56%	29	11 of 29	2 of 29
word sample	A′ (5 books, 1.95 M)	6-symbol (as §6.4)	364	22 of 50 (44%)	31–58%	28	9 of 28	2 of 28
word sample	A′ (5 books, 1.95 M)	6-symbol (as §6.4)	546	28 of 50 (56%)	42–69%	22	13 of 22	5 of 22
word sample	B (6 books, 4.67 M)	6-symbol (as §6.4)	182	10 of 50 (20%)	11–33%	40	33 of 40	8 of 40
word sample	B (6 books, 4.67 M)	6-symbol (as §6.4)	273	13 of 50 (26%)	16–40%	37	27 of 37	10 of 37
word sample	B (6 books, 4.67 M)	6-symbol (as §6.4)	364	19 of 50 (38%)	26–52%	31	23 of 31	7 of 31
word sample	B (6 books, 4.67 M)	6-symbol (as §6.4)	546	30 of 50 (60%)	46–72%	20	14 of 20	2 of 20
Wells prose	A′ (5 books, 1.95 M)	6-symbol (as §6.4)	182	9 of 50 (18%)	10–31%	41	29 of 41	7 of 41
Wells prose	A′ (5 books, 1.95 M)	6-symbol (as §6.4)	273	14 of 50 (28%)	17–42%	36	28 of 36	7 of 36
Wells prose	A′ (5 books, 1.95 M)	6-symbol (as §6.4)	364	14 of 50 (28%)	17–42%	36	24 of 36	7 of 36
Wells prose	A′ (5 books, 1.95 M)	6-symbol (as §6.4)	546	25 of 50 (50%)	37–63%	25	15 of 25	7 of 25
Wells prose	B (6 books, 4.67 M)	6-symbol (as §6.4)	182	6 of 50 (12%)	6–24%	44	32 of 44	6 of 44
Wells prose	B (6 books, 4.67 M)	6-symbol (as §6.4)	273	13 of 50 (26%)	16–40%	37	28 of 37	12 of 37
Wells prose	B (6 books, 4.67 M)	6-symbol (as §6.4)	364	14 of 50 (28%)	17–42%	36	27 of 36	10 of 36
Wells prose	B (6 books, 4.67 M)	6-symbol (as §6.4)	546	22 of 50 (44%)	31–58%	28	23 of 28	5 of 28

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Plaintext	Corpus (letters)	Score	Length	Truth kept	95% CI	Higher-scoring found	other corpus	fresh passage
Wells prose	A′ (5 books, 1.95 M)	chain-rule, 5 before	182	6 of 50 (12%)	6–24%	44	26 of 44	8 of 44
Wells prose	A′ (5 books, 1.95 M)	chain-rule, 5 before	273	12 of 50 (24%)	14–37%	38	20 of 38	7 of 38
Wells prose	A′ (5 books, 1.95 M)	chain-rule, 5 before	364	14 of 50 (28%)	17–42%	36	21 of 36	4 of 36
Wells prose	A′ (5 books, 1.95 M)	chain-rule, 5 before	546	15 of 50 (30%)	19–44%	35	21 of 35	11 of 35
Wells prose	B (6 books, 4.67 M)	chain-rule, 5 before	182	8 of 50 (16%)	8–29%	42	28 of 42	7 of 42
Wells prose	B (6 books, 4.67 M)	chain-rule, 5 before	273	14 of 50 (28%)	17–42%	36	23 of 36	11 of 36
Wells prose	B (6 books, 4.67 M)	chain-rule, 5 before	364	16 of 50 (32%)	21–46%	34	19 of 34	6 of 34
Wells prose	B (6 books, 4.67 M)	chain-rule, 5 before	546	25 of 50 (50%)	37–63%	25	15 of 25	5 of 25
Wells prose	A′ + B (6.62 M)	chain-rule, 3 before	182	4 of 50 (8%)	3–19%	46	—	4 of 46
Wells prose	A′ + B (6.62 M)	chain-rule, 3 before	273	5 of 50 (10%)	4–21%	44	—	9 of 44
Wells prose	A′ + B (6.62 M)	chain-rule, 3 before	364	8 of 50 (16%)	8–29%	42	—	11 of 42
Wells prose	A′ + B (6.62 M)	chain-rule, 3 before	546	15 of 50 (30%)	19–44%	35	—	7 of 35
Wells prose	A′ + B (6.62 M)	chain-rule, 4 before	182	4 of 50 (8%)	3–19%	46	—	11 of 46
Wells prose	A′ + B (6.62 M)	chain-rule, 4 before	273	7 of 50 (14%)	7–26%	43	—	10 of 43
Wells prose	A′ + B (6.62 M)	chain-rule, 4 before	364	10 of 50 (20%)	11–33%	40	—	9 of 40
Wells prose	A′ + B (6.62 M)	chain-rule, 4 before	546	14 of 50 (28%)	17–42%	36	—	3 of 36
Wells prose	A′ + B (6.62 M)	chain-rule, 5 before	182	5 of 50 (10%)	4–21%	45	—	16 of 45
Wells prose	A′ + B (6.62 M)	chain-rule, 5 before	273	17 of 50 (34%)	22–48%	33	—	8 of 33

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Plaintext	Corpus (letters)	Score	Length	Truth kept	95% CI	Higher-scoring found	other corpus	fresh passage
Wells prose	A′ + B (6.62 M)	chain-rule, 5 before	364	16 of 50 (32%)	21–46%	34	—	8 of 34
Wells prose	A′ + B (6.62 M)	chain-rule, 5 before	546	18 of 50 (36%)	24–50%	32	—	6 of 32
Wells prose	A′ + B (6.62 M)	6-symbol (as §6.4)	182	8 of 50 (16%)	8–29%	42	—	14 of 42
Wells prose	A′ + B (6.62 M)	6-symbol (as §6.4)	273	12 of 50 (24%)	14–37%	38	—	13 of 38
Wells prose	A′ + B (6.62 M)	6-symbol (as §6.4)	364	14 of 50 (28%)	17–42%	36	—	10 of 36
Wells prose	A′ + B (6.62 M)	6-symbol (as §6.4)	546	15 of 50 (30%)	19–44%	35	—	6 of 35

Data splits. The 13 corpus novels, sorted by file name, are split alternately: corpus A takes the 1st, 3rd, … (7 books), corpus B the 2nd, 4th, … (6 books); Section 6.4 uses A and B in full. For Section 6.5 the two Wells novels (*The Time Machine*, *The War of the Worlds*, both in A) are removed from every model and used only as plaintext (407,678 letters): A′ is A without them. The word sample is `lab/cipher/grille/plain_pool.txt`.

True tables. For each practice cipher the true table is the balanced table of the model's own corpus (letters packed into 13 groups by frequency, `poly_engine.balanced_table`), with its 13 labels shuffled by the plant's seed.

Search and score. Steepest ascent from the true table: every swap of two letters in different groups is scored, the best is taken, until none improves (at most 8 sweeps). The score is the mean log10 of the six-symbol pattern frequency over the passage (§6.4), or the mean log10 conditional probability of each symbol given the 3, 4 or 5 before it with add-one smoothing (chain-rule). A candidate table is scored by rebuilding the model from the corpus mapped through that table.

Seeds. Section 6.4: plant *p* uses `Random(9000 + p)` for its table labels and passage (the 20-plant runs `Random(7000 + p)`). Section 6.5: table labels `Random(12000 + p)`, passage start `Random(12500 + p)`, fresh passage 3,000 letters later. Four-square (§7): passage `Random(31000 + w)`, random-key controls `Random(55000 + w)`, order-test shuffles `Random(4242)`. GPU sums are not bit-for-bit deterministic, so individual runs can differ slightly on re-execution.

Appendix F. Language corpora and data provenance

Every external language source used in an experiment reported here, with where it came from, how it was processed and where it was used [F80]. Project Gutenberg texts were retrieved on October 11, 2026 from `https://www.gutenberg.org/cache/epub/{id}/pg{id}.txt`; the text between Gutenberg's `*** START OF` and `*** END OF` markers was kept (the license header and footer removed) and nothing else changed. Catalog details below are quoted from each book's Project Gutenberg record (read October 16, 2026); every last-update date the records

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list predates the retrieval (two Latin records list none). SHA-256 fingerprints of the files used are in `lab/cipher/lm/CORPUS.md`. A source's role can differ between experiments: in particular the two Wells novels are part of model corpus A in Section 6.4 but are held out of every model, as plaintext, in Section 6.5.

Table F.1. English novels (Project Gutenberg editions).

File	Work	Gutenberg	Record: released / last update	§6.4 corpus	§6.5 role	In repository
`adventures-holmes.txt`	*The Adventures of Sherlock Holmes*, Arthur Conan Doyle	#1661	Mar 1, 1999 / Oct 10, 2023	A	model corpus A′	yes
`great-expectations.txt`	*Great Expectations*, Charles Dickens (the text identifies itself as the 1867 edition)	#1400	Jul 1, 1998 / Dec 17, 2024	B	model corpus B	yes
`hound-baskervilles.txt`	*The Hound of the Baskervilles*, Arthur Conan Doyle	#2852	Oct 1, 2001 / Aug 4, 2025	A	model corpus A′	yes
`man-who-was-thursday.txt`	*The Man Who Was Thursday: A Nightmare*, G. K. Chesterton	#1695	Apr 1, 1999 / Feb 5, 2024	B	model corpus B	yes
`memoirs-holmes.txt`	*The Memoirs of Sherlock Holmes*, Arthur Conan Doyle	#834	Mar 1, 1997 / Dec 2, 2023	A	model corpus A′	yes
`mysterious-affair-styles.txt`	*The Mysterious Affair at Styles*, Agatha Christie	#863	Mar 1, 1997 / Apr 26, 2025	B	model corpus B	no — public domain in the USA, in copyright in the UK and EU
`return-holmes.txt`	*The Return of Sherlock Holmes*, Arthur Conan Doyle	#108	Mar 8, 2006 / Apr 25, 2026	A	model corpus A′	yes
`secret-adversary.txt`	*The Secret Adversary*, Agatha Christie	#1155	Jan 1, 1998 / Oct 29, 2022	B	model corpus B	no — public domain in the USA, in copyright in the UK and EU
`secret-agent.txt`	*The Secret Agent: A Simple Tale*, Joseph Conrad (the text is the Methuen second edition (October 1907))	#974	Jul 1, 1997 / Feb 12, 2023	A	model corpus A′	yes
`tale-of-two-cities.txt`	*A Tale of Two Cities*, Charles Dickens	#98	Jan 1, 1994 / Feb 26, 2026	B	model corpus B	yes

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File	Work	Gutenberg	Record: released / last update	§6.4 corpus	§6.5 role	In repository
`time-machine.txt`	*The Time Machine*, H. G. Wells	#35	Oct 2, 2004 / Jun 16, 2026	A	held-out plaintext (Wells prose)	yes
`war-and-peace.txt`	*War and Peace*, Leo Tolstoy, translated by Louise Maude and Aylmer Maude	#2600	Apr 1, 2001 / Jun 14, 2022	B	model corpus B	yes
`war-of-the-worlds.txt`	*The War of the Worlds*, H. G. Wells	#36	Oct 1, 2004 / Mar 30, 2026	A	held-out plaintext (Wells prose)	yes

Also requested on October 11 but never retrieved, because Gutenberg returned server errors: eBooks 558, 2097 and 1342. They are not in any corpus.

Table F.2. Which text was used where.

Text	Size (letters)	Built from	Used in (this report)	Reproduce
Model text	7,032,556	all 13 novels, letters A–Z only, upper case, in file-name order	§4 example table (its first 2,000,000 letters: *Adventures*, *Great Expectations*, *Hound*, *The Man Who Was Thursday* and part of *Memoirs*); §6.1 [F78], §6.2 [F74], §6.3 [I47]	`tools/build_model_text.py` (needs the two Christie novels)
Corpus A / B	2,360,321 / 4,672,235	the 13 novels in file-name order, alternately (Table F.1)	§6.4 [F76]	`round11/identifiability.py`
Corpus A′ / B / A′ + B	1,952,643 / 4,672,235 / 6,624,878	A without the Wells novels; B; both	§6.5 [F77, F79]	`round12/review_checks.py`, `round12/comparability.py`

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Text	Size (letters)	Built from	Used in (this report)	Reproduce
Wells prose	407,678	*The Time Machine* + *The War of the Worlds*, held out of every model in these experiments	§6.5 plaintexts and fresh passages [F77], §6.5 comparability [F79], §7 purpose-built four-square passages [O34]	as above; `round12/review_checks.py fsq`
Word sample	60,000	English words drawn by frequency from wordfreq (below); characters 130,000–190,000 of the stream	practice plaintexts of §6.1–6.4; word-sample setups of §6.5; §7 random-key four-square [R35]; [F71]	`lab/cipher/grille/plain_pool.txt`
Quadgram table	26^4 values	four-letter frequencies of the same wordfreq stream	[F71] (grille search with the letter key given)	`lab/cipher/grille/quad_en.npy`
English word stream	200,000 per call	the same wordfreq sampling, generated on the fly	§2 flatness comparison [F31]; the book's sign system [R31]	PegasusDMZ agent tools (not in this repository)
Latin corpus	≈ 2.7 M after filtering	the 10 Latin editions of Table F.3; lines that are mostly English removed	§2 Latin flatness comparison [F31, F80]	`lab/cipher/provenance/latin_flatness.py` (needs the agent tools)

Plaintext and language model never come from the same text within an experiment: the word sample is drawn from wordfreq, not from the novels, and the Section 6.5 checks remove the Wells novels from every model before using them as plaintext.

Table F.3. Latin editions (Project Gutenberg), used only for the Latin comparison of Section 2. Not redistributed; fingerprints of the files used are in `lab/cipher/lm/CORPUS.md`.

File	Work	Gutenberg	Record language	English apparatus
`augustine-confessiones.txt`	*Confessiones*, Augustine of Hippo; ed. Philippe Goibaud du Bois; E. B. Pusey	#33849	Latin	no

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File	Work	Gutenberg	Record language	English apparatus
`caesar-bello-gallico.txt`	*C. Iuli Caesaris De Bello Gallico, I-IV*, Julius Caesar	#218	Latin	no
`cicero-catilina.txt`	*Speeches against Catilina*, Marcus Tullius Cicero; ed. Ernest Alfred Upcott	#39355	English; Latin	yes — English notes or translation
`cicero-de-officiis.txt`	*De Officiis*, Marcus Tullius Cicero; Walter Miller (translator)	#47001	English	yes — English notes or translation
`cicero-orations.txt`	*Cicero's Orations*, Marcus Tullius Cicero	#226	Latin	no
`descartes-meditationes.txt`	*Meditationes de prima philosophia*, René Descartes	#23306	Latin	no
`gellius.txt`	*Stories from Aulus Gellius: Being Selections and Adaptations from the Noctes Atticae*, Aulus Gellius; ed. G. Herbert Nall	#25861	Latin	yes — English notes or translation
`newton-principia.txt`	*Philosophiae Naturalis Principia Mathematica*, Isaac Newton	#28233	Latin	no
`virgil-aeneid.txt`	*Aeneidos*, Virgil	#227	Latin	no
`viri-romae.txt`	*Selections from Viri Romae*, C. F. L'Homond; ed. Robert Arrowsmith; Charles Knapp	#33311	English; Latin	yes — English notes or translation

The four editions with English apparatus are school editions or a facing translation; the Latin reader drops every line in which English function words make up 20% or more of the words, which leaves 0.5% English words in the stream. The Latin result of Section 2 depends on which editions are sampled: 345–374 of 1,000 windows for the full corpus, 167–182 for the six editions without English apparatus [F80].

Word frequencies. The word sample, the quadgram table and the English word stream come from the `wordfreq` package (Speer 2022; version 3.1.1 installed — regenerating the two shipped files with it reproduces them byte for byte): the 30,000 most frequent English words of its default list, sampled in proportion to frequency with a fixed seed (4321) and joined without spaces. wordfreq's English list combines Wikipedia, subtitles (OPUS OpenSubtitles 2018 and SUBTLEX: Brysbaert and New 2009; van Heuven et al. 2014), news, Google Books Ngrams, web text, Twitter and Reddit. Its code is under the Apache License 2.0 and its data files under CC BY-SA 4.0; the derived files in this repository are shared under CC BY-SA 4.0 accordingly (see `NOTICE`), not under the repository's CC BY 4.0.

Rights. Project Gutenberg's records state each book is public domain in the USA. The two Agatha Christie novels (author died 1976) remain in copyright in the UK and the EU, so they, and the model text built from them, are not redistributed. For the other English novels the authors and the War and Peace translators died before 1955; the Latin edi-

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